

Möbius Energy and the Light-Plane Correspondence

Exact representations and a conditional critical-line theorem

Christophe Michaels

6 October 2026

Abstract

We give a unified analytical treatment of the Green energy of the finite Möbius source. Its cumulative, block, field, finite spectral, circle, and Hardy-space representations preserve the complete quadratic form, including inherited interactions and terminal contributions. The compensated Möbius modifier yields exact decompositions for completed and unfinished excursions. A boundary projection has uniformly bounded discarded energy. Three-step grouping with one-step boundary return has a uniform complete-energy cost on each sixfold interval. A shared projection pays the error of an entire scale chain. Completed multiplicative families admit a sixth-root comparison. Two circle phases return the line energy exactly; complete derivative data and their quadratic lift admit faithful Hilbert-space realizations with controlled transfer cost. Under an explicitly stated bound for the complete signed scale increment, a finite descent gives subpower energy. The terminal term then yields the Mertens estimate needed for Mellin holomorphy, zero exclusion, and functional-equation reflection. The scale-increment bound is an assumption of this theorem. The manuscript consolidates the existing analytical proofs; it contains no numerical experiments or numerical extrapolation.

Keywords. Möbius function; Green energy; Mellin transform; arithmetic Hilbert spaces; boundary compression; Hardy space; Riemann hypothesis.

Contents

1	The arithmetic observable and the main theorem	2
2	The Green form and compensated arithmetic blocks	3
3	Records, long intervals, and terminal recovery	10
4	The field, source Hamiltonian, and horizon observable	12
5	Exact prefix carry and completed-prefix mode estimates	16
6	Orthogonal boundary compression and one chain allowance	18
7	Completed multiplicative families and their signed complement	20
8	Positive circle domains and exact energy return	24
9	The complete derivative record in a fixed Hardy space	26

10	A calibrated infinite derivative lift and its quadratic readout	29
11	Finite descent and the Mellin conclusion	32
A	The centered scalar comparison and its compression	34
B	Prime-discrepancy blocks and their complete Mellin return	38
C	Logical scope, normalization, and source concordance	40

1 The arithmetic observable and the main theorem

All logarithms are natural. Real arithmetic cutoffs are interpreted by their integer parts. The Möbius function is

$$\mu(1) = 1, \quad \mu(n) = (-1)^{\omega(n)} \quad \text{if } n \text{ is squarefree,} \quad \mu(n) = 0 \quad \text{otherwise,}$$

where $\omega(n)$ counts distinct prime factors. Set

$$M(x) = \sum_{n \leq x} \mu(n), \quad h(x) = \sum_{n \leq x} \frac{\mu(n)}{n}, \quad D_N(s) = \sum_{n \leq N} \mu(n) n^{-s}. \quad (1.1)$$

The complete energy of the prefix through N is

$$R(N) = \sum_{a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)}, \quad R(0) = 0. \quad (1.2)$$

Its positive cumulative representation, proved below, is

$$R(N) = I_M(N) + \frac{M(N)^2}{N}, \quad I_M(N) = \int_1^N \frac{M(x)^2}{x^2} dx = \sum_{k < N} \frac{M(k)^2}{k(k+1)}. \quad (1.3)$$

The terminal $M(N)^2/N$ is part of the observable.

For $X \geq 1$ define its nondecreasing envelope by

$$S(X) = \max\{1, R(n) : 1 \leq n \leq \lfloor X \rfloor\}, \quad \mathcal{B}(X) = 1 + S(X). \quad (1.4)$$

For $N \geq 6$, write

$$L_N = \lfloor N/6 \rfloor, \quad \delta_N = R(N) - R(L_N), \quad \mathfrak{c}_N = \frac{[\delta_N]_+}{\mathcal{B}(N^{1/6})^5}, \quad \alpha = \frac{\log 5}{\log 6} < 1. \quad (1.5)$$

The positive part is taken after the complete signed increment has been formed. In particular, it is not taken separately on its inherited interaction and its tail energy.

Hypothesis 1.1 (Complete scale-increment allowance). There are constants $C > 0$, $A \geq 0$, $a \geq 0$ and $0 < \rho < 1$, independent of N , such that, for every integer $N \geq 6$,

$$\delta_N \leq C(1 + \log N)^a \exp\{A(\log N)^\rho\} \mathcal{B}(N^{1/6})^5. \quad (1.6)$$

The constant-coefficient version is $A = a = 0$. This is the quantitative premise of the light-plane growth framework. The exact representations below specify its arithmetic image and preserve its entire numerator.

Theorem 1.2 (Conditional critical-line theorem). *Assume Hypothesis 1.1. Then*

$$\log \mathcal{B}(X) = \begin{cases} O((\log X)^\alpha), & \rho < \alpha, \\ O((\log X)^\alpha \log(2 + \log X)), & \rho = \alpha, \\ O((\log X)^\rho), & \rho > \alpha. \end{cases} \quad (1.7)$$

When $A = 0$, the first bound holds independently of ρ . Consequently, for every $\varepsilon > 0$,

$$R(N) = O_\varepsilon(N^\varepsilon), \quad M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \quad (1.8)$$

The function

$$F(s) = s \int_1^\infty M(x) x^{-s-1} dx \quad (1.9)$$

is holomorphic on $\operatorname{Re} s > 1/2$ and agrees with $1/\zeta(s)$ on $\operatorname{Re} s > 1$. Every nontrivial zero of ζ has real part $1/2$.

The proof is completed in Section 11. The preceding sections assemble the exact arithmetic construction and its geometric representations. The same inequality can be read as a signed trace in the quadratic plane, as a difference of Hardy norms, or as a difference of weighted circle energies. These are readings of one hypothesis. They are not successive additional hypotheses.

1.1 Organization and conventions

Physical charges a_n and weighted coefficients $c_n = a_n/n$ are kept distinct. The two matrix kernels are related by

$$\frac{1}{\max(a, b)} = \frac{\min(a, b)}{ab}, \quad \mathbf{B}_N(a, b) = \min(a, b). \quad (1.10)$$

All complex inner products are conjugate-linear in their first argument. Finite-dimensional coefficient spaces are identified with their zero-extended images at larger cutoffs. A norm with subscript $\mathbf{B}$ means $\|c\|_{\mathbf{B}}^2 = c^* \mathbf{B}_N c$.

The term *light plane* denotes the Hilbert-space correspondence defined in Sections 4–10, within the arithmophysics framework. Its static field operator, source Hamiltonian, circle domains, and projections are specified mathematical objects. The *infinity plane* retains the complete derivative record; the *quadratic plane* retains its full outer product. Their use here is through explicit norm identities and bounded maps.

The statements and proofs are consolidated from the running manuscript [2], the modifier and record papers [3, 4], the lifetime continuation [5], and the subsequent horizon and arithmophysics papers [6, 8, 9, 10, 11, 12, 13, 14, 15]. Source locations are given in Appendix C. The analytical statements have been reordered and notation has been standardized. The scale allowance (1.6) retains its status as a premise in those sources. No claim that it follows from the coordinate identifications alone is used in the theorem.

2 The Green form and compensated arithmetic blocks

The results of this section consolidate the native metric and modifier of [3, Sections 2–11], together with the inherited-interaction formula of [5, Sections 92–95].

Theorem 2.1 (Cumulative representation). *Let $a, b \in \mathbb{C}^N$, and put $A(k) = \sum_{n \leq k} a_n$ and $B(k) = \sum_{n \leq k} b_n$. Then*

$$\sum_{m, n \leq N} \frac{\overline{a_m} b_n}{\max(m, n)} = \sum_{k < N} \frac{\overline{A(k)} B(k)}{k(k+1)} + \frac{\overline{A(N)} B(N)}{N}. \quad (2.1)$$

The form is positive definite and is unchanged by extending the ambient cutoff while keeping the charges fixed. If $A_N(x) = \sum_{n \leq \min(x, N)} a_n$, then

$$\|a\|_G^2 := \sum_{m, n \leq N} \frac{\overline{a_m} a_n}{\max(m, n)} = \int_1^\infty \frac{|A_N(x)|^2}{x^2} dx. \quad (2.2)$$

Proof. For $m, n \leq N$,

$$\frac{1}{\max(m, n)} = \frac{1}{N} + \sum_{k=\max(m, n)}^{N-1} \frac{1}{k(k+1)}.$$

Insert this telescoping identity into the finite double sum. The coefficient of $1/[k(k+1)]$ is $\overline{A(k)} B(k)$. If the norm vanishes, all cumulative sums vanish, and hence all charges vanish. The kernel entries are independent of the ambient cutoff. Integrating the step function on $[k, k+1)$, and its constant terminal value on $[N, \infty)$, proves (2.2). $\square$

Corollary 2.2 (Successor and interval identities). *The Möbius energy obeys*

$$R(n) - R(n-1) = \frac{M(n)^2 - M(n-1)^2}{n} = \frac{2\mu(n)M(n-1) + \mu(n)^2}{n}, \quad (2.3)$$

$$0 \leq R(n) \leq 2n - H_n, \quad H_n = \sum_{k \leq n} \frac{1}{k}. \quad (2.4)$$

For integers $1 \leq m < n$,

$$R(n) - R(m) = \frac{M(n)^2 - M(m)^2}{n} + \sum_{k=m}^{n-1} \frac{M(k)^2 - M(m)^2}{k(k+1)}. \quad (2.5)$$

Proof. Collect the pairs with maximum index n in (1.2) to obtain (2.3). Bounding each charge in absolute value by one gives $\sum_{k \leq n} (2k-1)/k = 2n - H_n$. Subtract the cumulative formulas at m, n and use $\sum_{k=m}^{n-1} 1/[k(k+1)] = 1/m - 1/n$. $\square$

2.1 The modifier and its exact compensation

Let $t_0 < t_1 < \dots < t_r = N$ be an integer partition. For a charge vector a supported on $(t_0, N]$, define

$$\mathcal{M}_{\mathcal{P}} a = \sum_{j=1}^r q_j e_{t_j}, \quad q_j = \sum_{t_{j-1} < n \leq t_j} a_n, \quad \eta_{\mathcal{P}} = a - \mathcal{M}_{\mathcal{P}} a. \quad (2.6)$$

This is the endpoint form of the Michaels Möbius modifier. For a fixed partition it is an algebraic projection. Its exact residual is part of the reconstruction.

Proposition 2.3 (Endpoint compensation). *The modifier preserves every cumulative sum at a partition endpoint and satisfies $\mathcal{M}_{\mathcal{P}}^2 = \mathcal{M}_{\mathcal{P}}$. With $A_j(k) = \sum_{t_{j-1} < n \leq k} a_n$, its residual has energy*

$$\|\eta_{\mathcal{P}}\|_G^2 = \sum_{j=1}^r \sum_{k=t_{j-1}+1}^{t_j-1} \frac{|A_j(k)|^2}{k(k+1)}. \quad (2.7)$$

If $H_j = \max_{t_{j-1} < k < t_j} |A_j(k)|$, with $H_j = 0$ for a singleton cell, then

$$\|\eta_{\mathcal{P}}\|_G^2 \leq \sum_j H_j^2 \left(\frac{1}{t_{j-1} + 1} - \frac{1}{t_j} \right). \quad (2.8)$$

Proof. An endpoint belongs to its own cell, so applying the modifier twice has the same effect as applying it once. At every endpoint the residual has cumulative sum zero. Within a cell its cumulative sum is $A_j(k)$, since the endpoint compensation has not yet occurred. The terminal is zero. Theorem 2.1 proves (2.7); bounding each local cumulative sum and telescoping proves (2.8). $\square$

The projection in (2.6) need not be orthogonal. The energy identity is therefore

$$\|a\|_G^2 = \|\mathcal{M}_{\mathcal{P}}a\|_G^2 + 2\operatorname{Re}\langle \mathcal{M}_{\mathcal{P}}a, \eta_{\mathcal{P}} \rangle_G + \|\eta_{\mathcal{P}}\|_G^2.$$

The orthogonal replacement in Section 6 preserves two moments, which removes this mixed term in the full metric.

2.2 Completed blocks and arbitrary unfinished tails

Theorem 2.4 (Future orthogonality). *Let a be supported on $(u, v]$. For every $j \geq v$,*

$$\langle a, e_j \rangle_G = \frac{1}{j} \overline{\sum_{u < n \leq v} a_n}. \quad (2.9)$$

If at least one later coordinate is available, zero total charge is equivalent to orthogonality to every vector supported after v . Consequently successive completed zero-charge blocks are mutually orthogonal, and are orthogonal to a final unfinished block. Their energies add exactly.

Proof. For $n \leq v \leq j$ the common denominator is j . This proves (2.9), its converse, and the statement for arbitrary later vectors by linearity. Expanding the square of a sum of successive blocks proves exact addition. $\square$

Fix an initial integer $t \geq 0$. Partition the Möbius sequence after t by successive first returns to the level $M(t)$, retaining a final unfinished interval at any observation cutoff N . Zero singleton blocks are permitted. Within a nontrivial first-return block $B = (u, v]$, the relative cumulative sum

$$s_B(k) = M(k) - M(t) = \sum_{u < n \leq k} \mu(n)$$

has a single strict sign before closure: a walk with increments in $\{-1, 0, 1\}$ cannot change sign without returning to zero.

For a completed block and for the unfinished block, respectively,

$$E_B = \sum_{k=u+1}^{v-1} \frac{s_B(k)^2}{k(k+1)}, \quad (2.10)$$

$$E_{(u, N]} = \sum_{k=u+1}^{N-1} \frac{s_B(k)^2}{k(k+1)} + \frac{s_B(N)^2}{N}. \quad (2.11)$$

Theorem 2.4 gives the native budget

$$\|\mu \mathbf{1}_{(t,N]}\|_G^2 = \sum_{B \text{ completed}} E_B + E_{\text{open}}. \quad (2.12)$$

In particular $t = 0$ gives the complete energy $R(N)$.

The smoothing of the adaptive endpoint output can be read exactly. Starting at $t = 0$, let a_* be the last completed return before N , and let $\mathcal{P}$ include the unfinished cell ending at N when needed. Then $M(a_*) = 0$ and

$$\mathcal{M}_{\mathcal{P}} \mu_{\leq N} = M(N) e_N, \quad \|\mathcal{M}_{\mathcal{P}} \mu_{\leq N}\|_G^2 = \frac{M(N)^2}{N}. \quad (2.13)$$

The residual has total charge zero, so it is orthogonal to e_N . Consequently the original modifier identity gives

$$R(N) = \frac{M(N)^2}{N} + \|\eta_{\mathcal{P}}\|_G^2, \quad \|\eta_{\mathcal{P}}\|_G^2 = I_M(N). \quad (2.14)$$

This is the exact correspondence between the endpoint smoothing and the historical energy [18, Section 6.3]. At a completed return the endpoint output vanishes; every completed block energy is retained in the residual. For these adaptive cells, their actual heights enter (2.8). The width-three estimate (2.26) applies when the chosen widths are at most three.

Proposition 2.5 (Intrinsic block bounds). *For a completed block of height H_B and length $\ell_B = v - u$,*

$$E_B \leq H_B^2 \left(\frac{1}{u+1} - \frac{1}{v} \right), \quad (2.15)$$

$$|s_B(u+j)| \leq \min\{j, \ell_B - j, H_B\}, \quad 1 \leq j < \ell_B. \quad (2.16)$$

If $Q_u(k) = \sum_{u < n \leq k} \mu(n)^2$ and $Q_B = Q_u(v)$, then

$$|s_B(k)| \leq \min\{Q_u(k), Q_B - Q_u(k), H_B\}. \quad (2.17)$$

For an unfinished block, $E_{(u,N]} \leq H_B^2/(u+1)$.

Proof. Bound the cumulative squares in (2.10) and telescope. A path has made at most j unit steps at $u+j$ and must cancel its current sum within $\ell_B - j$ remaining unit steps. These facts prove (2.16). Because $|\mu(n)| = \mu(n)^2$, the prefix and suffix squarefree counts give (2.17). In the open case the weights, including the terminal, sum to $1/(u+1)$. $\square$

An equivalent occupation formula retains every duration. If

$$\Lambda_B(h) = \sum_{k=u+1}^{v-1} \frac{\mathbf{1}_{|s_B(k)| \geq h}}{k(k+1)},$$

then

$$E_B = \sum_{h=1}^{H_B} (2h-1) \Lambda_B(h). \quad (2.18)$$

Indeed $s^2 = \sum_{h=1}^{|s|} (2h-1)$; insert this identity and interchange finite sums. The open version includes $\mathbf{1}_{|s_B(N)| \geq h}/N$ in $\Lambda_B(h)$.

2.3 Signed work and the inherited total

Proposition 2.6 (Signed work identity). *For either a completed block ending at v or an unfinished block observed at v ,*

$$E_B = \sum_{n=u+1}^v \frac{\mu(n)^2 + 2s_B(n-1)\mu(n)}{n}, \quad (2.19)$$

where the energy includes its terminal when it is unfinished.

Proof. The numerator is $s_B(n)^2 - s_B(n-1)^2$. Summation by parts with weight $1/n$ yields (2.10) or (2.11), according to whether $s_B(v) = 0$. $\square$

Write $A = M(t)$, and for each nonzero block put

$$h_B = \sum_{n \in B} \frac{\mu(n)}{n}, \quad H_B = \max |s_B|, \quad \sigma_B = \text{sign}(s_B).$$

For completed blocks define

$$V_B = \sum_{k=u+1}^{v-1} \frac{|s_B(k)|(H_B - |s_B(k)|)}{k(k+1)}.$$

For the open block add $|s_B(N)|(H_B - |s_B(N)|)/N$ and end its historical sum at $N-1$.

Theorem 2.7 (Complete inherited-interaction budget). *For every finite cutoff,*

$$R(N) = R(t) + \sum_B (E_B + 2Ah_B) \quad (2.20)$$

$$= R(t) + \sum_B [(H_B + 2\sigma_B A)|h_B| - V_B], \quad V_B \geq 0. \quad (2.21)$$

If $\sigma_B A = -|A|$ and $H_B \leq 2|A|$, that block's complete signed contribution is nonpositive, independently of its length.

Proof. Completed blocks have zero total, so all interactions with subsequent blocks vanish. Their interaction with the fixed original prefix is Ah_B . This proves (2.20). Summation by parts gives

$$h_B = \sum_{k=u+1}^{v-1} \frac{s_B(k)}{k(k+1)}$$

for a completed block, with $s_B(N)/N$ added for an open block. Because the cumulative sum has constant sign, $h_B = \sigma_B |h_B|$. Termwise subtraction shows $E_B = H_B |h_B| - V_B$. This proves (2.21) and its final assertion. $\square$

2.4 Weighted transfer without a block-count loss

Proposition 2.8 (Full-prefix transfer). *For an absolutely continuous real or complex probe F on $[1, N]$,*

$$\left| \sum_{n \leq N} \mu(n) F(n) \right|^2 \leq R(N) \left\{ \int_1^N t^2 |F'(t)|^2 dt + N |F(N)|^2 \right\}. \quad (2.22)$$

For real probes $F_1, \dots, F_r$, write $y_i = \sum_{n \leq N} \mu(n) F_i(n)$ and

$$G_{ij} = \int_1^N t^2 F'_i(t) F'_j(t) dt + N F_i(N) F_j(N).$$

Then $yy^\top \preceq R(N)G$.

Proof. Abel summation gives

$$\sum_{n \leq N} \mu(n) F(n) = M(N)F(N) - \int_1^N M(t)F'(t) dt.$$

Apply Cauchy–Schwarz to the direct sum of $M(t)/t \in L^2[1, N]$ and $M(N)/\sqrt{N}$, paired with $-tF'(t)$ and $\sqrt{N}F(N)$. Their first squared norm is exactly $R(N)$. Apply the scalar inequality to every real linear combination of the F_i to obtain the matrix inequality. $\square$

The budget in (2.12) can be substituted for $R(N)$ in (2.22). This uses the aggregate completed-plus-open energy once. It introduces no factor equal to the number of blocks and assumes no orthogonality after applying the probes.

2.5 Three-step specialization and boundary return

The compensated three-step construction already appears in the September 8 edition [18, Sections 3–6 and Corollary 4.4]. That edition defines the exceptional 2, 3 modifier, endpoint compression with exact compensation, and the residual bound $4/(L+1)$ for blocks of length at most three. These are existing results; the date identifies the available edition of the earlier construction. They are also retained in the expanded edition [3, Sections 3–4]. The underlying modifier allows arbitrary partitions. Its dynamic DNA version uses variable first-return blocks, as in Section 2.2; the original source treats both constructions. The bounded-width specialization below supplements that adaptive construction and does not restrict its possible block lengths. The later note [17, Sections 1–7] supplies the complete-energy comparison below and specifies an additional boundary convention: final widths 3, 1 are replaced by 2, 2. This convention separates the retained endpoints while the compensation retains the original coefficients. The later convention is not used as a definition of the historical three-step description.

Fix integers $L \geq 1$ and $N - L \geq 2$. Write $N - L = 3q + r$ with $r \in \{0, 1, 2\}$. Partition $(L, N]$ into q widths of three if $r = 0$; into q widths of three followed by two if $r = 2$; and into $q - 1$ widths of three followed by two and two if $r = 1$. Every width is two or three. Denote the endpoints by $L = t_0 < t_1 < \dots < t_m = N$.

For physical charges $a_n \in \mathbb{C}$, $|a_n| \leq 1$, set

$$\tilde{a}_{L,N} = \sum_{n \leq L} a_n e_n + \sum_{j=1}^m q_j e_{t_j}, \quad q_j = \sum_{t_{j-1} < n \leq t_j} a_n, \quad \eta = a - \tilde{a}_{L,N}. \quad (2.23)$$

Put $E(a) = \|a\|_G^2$. Write $A(u) = \sum_{n \leq u} a_n$ and $S_j(k) = \sum_{t_{j-1} < n \leq k} a_n$. The exact signed difference is

$$E(a) - E(\tilde{a}_{L,N}) = \sum_j \sum_{k=t_{j-1}+1}^{t_j-1} \frac{2 \operatorname{Re}\{\overline{A(t_{j-1})} S_j(k)\} + |S_j(k)|^2}{k(k+1)}. \quad (2.24)$$

This is the subtraction of the two cumulative energy formulas. Endpoint sums, including $A(N)$, agree exactly.

Theorem 2.9 (Complete three-step comparison; [17]). *For every such source and partition,*

$$|E(a) - E(\tilde{a}_{L,N})| \leq C_{\mathcal{P}}(L, N) \leq 2 \log(N/L), \quad (2.25)$$

$$C_{\mathcal{P}}(L, N) := \sum_j \sum_{k=t_{j-1}+1}^{t_j-1} \frac{k^2 - t_{j-1}^2}{k(k+1)}.$$

The original bounded-length residual estimate [18, Corollary 4.4] also gives

$$E(\eta) \leq 4 \left(\frac{1}{L+1} - \frac{1}{N} \right). \quad (2.26)$$

Both assertions are unconditional and include the entire earlier prefix.

Proof. On a block beginning at u , put $k = u + r$. The absolute numerator in (2.24) is at most $2ur + r^2 = k^2 - u^2$. The width-two and width-three allowances in the source proof are

$$D_2(u) = \frac{2u+1}{(u+1)(u+2)} \leq \frac{4}{u+2}, \quad D_3(u) = \frac{2u+1}{(u+1)(u+2)} + \frac{4u+4}{(u+2)(u+3)} \leq \frac{6}{u+3}.$$

For the latter, the difference between the upper bound and $D_3(u)$ is $(3u+5)/[(u+1)(u+2)(u+3)]$. Hence a block of width h costs at most $2h/(u+h) \leq 2 \int_u^{u+h} dx/x$. Summing gives (2.25). Every interior residual cumulative sum has modulus at most two. The residual identity (2.7) and telescoping over $L < k < N$ give (2.26). $\square$

For the whole-prefix partition, let $R_3(N)$ be the modified Möbius energy, with the singleton convention at $N = 1$. The first block begins at zero; its allowance is at most $7/6$. The remaining blocks satisfy Theorem 2.9. Thus

$$|R(N) - R_3(N)| \leq \frac{7}{6} + 2 \log N. \quad (2.27)$$

This supplies an unconditional transfer of an every-positive-exponent bound between the original and modified complete energies.

2.6 The exceptional adjacent pair and its compensation

The original finite modifier is

$$\mu_{\text{exc}}(n) = \mu(n) + \mathbf{1}_{n=2} + \mathbf{1}_{n=3}, \quad \mu_{\leq N} = (\mu_{\text{exc}})_{\leq N} - e_2 - e_3 \quad (N \geq 3). \quad (2.28)$$

It neutralizes the two exceptional prime coefficients. Their compensation has energy $E(e_2 + e_3) = 3/2$ [3, Proposition 3.2]. The complete energy comparison retained in [17, Section 5] is

$$R_{\text{exc}}(N) = R(N) + 4h(N) - \frac{1}{2} \quad (N \geq 3). \quad (2.29)$$

Indeed, each pairing $\langle \mu_{\leq N}, e_2 \rangle_G$ and $\langle \mu_{\leq N}, e_3 \rangle_G$ equals $h(N) - 1/2$; expand the square. The divisor identity

$$1 = \sum_{n \leq N} \mu(n) \lfloor N/n \rfloor = Nh(N) - \sum_{n \leq N} \mu(n) \{N/n\}$$

gives $|h(N)| \leq 1$. Consequently, for $N, L \geq 3$,

$$|[R_{\text{exc}}(N) - R_{\text{exc}}(L)] - [R(N) - R(L)]| \leq 8. \quad (2.30)$$

This modifier preserves the original energy through its explicit compensation. Prime-exponent labels such as 2^3 and 3^2 remain part of the valuation data; both of their Möbius coefficients are zero. The present observable is specified by those coefficients and the Green metric.

2.7 Return to the identical scale denominator

For $N \geq 6$ let $L = L_N = \lfloor N/6 \rfloor$ and apply (2.23) to the actual Möbius tail. The complete modified increment is

$$\mathcal{Q}_N^{(3)} := E(\tilde{\mu}_{L,N}) - R(L) = \sum_{i,j} \frac{q_i q_j}{\max(t_i, t_j)} + 2M(L) \sum_j \frac{q_j}{t_j}. \quad (2.31)$$

The compressed tail retains every block mass and its terminal. Its pairing with the unchanged prefix is the second displayed term. Since $N \leq 6L + 5 \leq 11L$, Theorem 2.9 gives

$$|\delta_N - \mathcal{Q}_N^{(3)}| \leq 2 \log(N/L) \leq 2 \log 11. \quad (2.32)$$

With the original denominator from (1.5), define

$$\hat{\mathbf{c}}_N^{(3)} = \frac{[\mathcal{Q}_N^{(3)}]_+}{\mathcal{B}(N^{1/6})^5}.$$

Then, unconditionally,

$$|\mathbf{c}_N - \hat{\mathbf{c}}_N^{(3)}| \leq \frac{2 \log(N/L)}{\mathcal{B}(N^{1/6})^5} \leq \frac{\log 11}{16}. \quad (2.33)$$

Here the positive-part map is 1-Lipschitz and $\mathcal{B} \geq 2$. Thus a finite upper bound on $\hat{\mathbf{c}}_N^{(3)}$ supplies the fixed-coefficient hypothesis of Theorem 11.2; conversely such a bound on $\mathbf{c}_N$ transfers to the modified coefficient. The polylogarithmic and stretched-logarithmic versions transfer as well. If the exceptional source is also used, its complete increment returns at cost at most $8 + 2 \log(N/L)$ for $L \geq 3$, by (2.30). The finitely many smaller horizons are absorbed into the base constant.

The field, Hardy, and quadratic representations below preserve the same complete energy. In the Hardy realization $\mathcal{U}$, for example, writing $f = \mathcal{U}\tilde{a}_{L,N}$ and $g = \mathcal{U}\eta$ retains

$$|f + g\rangle\langle f + g| = |f\rangle\langle f| + |f\rangle\langle g| + |g\rangle\langle f| + |g\rangle\langle g|.$$

Taking its trace returns the full compensated identity. Equations (2.32)–(2.33) bound the cost of modifying the source at every scale. The bound on the actual retained expression (2.31) is the quantitative hypothesis in these coordinates; [17, Section 7] leaves that estimate open.

3 Records, long intervals, and terminal recovery

This section records the analytical long-interval results of [4, Theorems 3.2–6.1]. To include the initial jump, put $S_0(X) = \max_{0 \leq n \leq \lfloor X \rfloor} R(n)$ for $X \geq 0$. Thus $S_0(0) = 0$ and $S_0(X) = S(X)$ for $X \geq 1$.

An integer $N \geq 1$ is a strict energy record if $R(N) > S_0(N - 1)$. Whenever attained, define

$$\tau_h = \min\{n \geq 0 : |M(n)| = h\}, \quad \tau_0 = 0.$$

Theorem 3.1 (Energy records are first passages). *If N is a strict energy record and $h = |M(N)|$, then $h \geq 1$, $N = \tau_h$, and $|M(k)| \leq h - 1$ for $k < N$. Distinct records have distinct positive integer heights. At a noninitial record, with $m = \tau_{h-1} \geq 1$,*

$$R(N) - R(m) = \frac{2h - 1}{N} - \sum_{k=m}^{N-1} \frac{(h - 1)^2 - M(k)^2}{k(k + 1)}. \quad (3.1)$$

Proof. If N is not a new absolute maximum of M , choose $t < N$ at which the maximum through N is attained. Every numerator in (2.5), with $m = t$, is nonpositive. Thus $R(N) \leq R(t)$, contradicting the record property. A unit-step integer walk must pass every smaller absolute height before attaining h , and must satisfy $|M(N-1)| = h-1$. Finally apply (2.5) with $m = \tau_{h-1}$. $\square$

Lemma 3.2 (Envelope jump). *For every $N \geq 1$,*

$$0 \leq \sqrt{S_0(N)} - \sqrt{S_0(N-1)} \leq N^{-1/2}. \quad (3.2)$$

At a record of height h , $S_0(N) - S_0(N-1) \leq (2h-1)/N$.

Proof. At a record, (2.3) and $S_0(N-1) \geq R(N-1)$ bound the difference of squares by $(2h-1)/N$. The terminal terms give $\sqrt{S_0(N)} \geq h/\sqrt{N}$ and $\sqrt{S_0(N-1)} \geq (h-1)/\sqrt{N}$. Divide the difference of squares by the sum of these lower bounds. For $N = 1$ the second bound is zero. Away from records the increment is zero. $\square$

Theorem 3.3 (Summable charge of long records). *For every $\beta > 1/2$, the noninitial records with $N - \tau_{|M(N)|-1} > N^\beta$ satisfy*

$$\sum_{\substack{N \text{ record}, N > 1 \\ N - \tau_{|M(N)|-1} > N^\beta}} (\sqrt{S_0(N)} - \sqrt{S_0(N-1)}) \leq \frac{1}{\beta - 1/2}. \quad (3.3)$$

Proof. Put $m = \tau_{|M(N)|-1}$. On a long record,

$$\Delta\sqrt{S_0(N)} \leq N^{-1/2} \leq \frac{N-m}{N^{\beta+1/2}} \leq \int_m^N t^{-\beta-1/2} dt.$$

The selected adjacent-level first-passage intervals are disjoint and lie in $[1, \infty)$. Sum and integrate over that half-line. $\square$

Lemma 3.4 (Mixed increment accounting). *Let s_j be nondecreasing with $s_0 = 0$, and partition its positive increments into classes A, D . Put*

$$a_n = \sum_{j \leq n, j \in A} (s_j - s_{j-1}), \quad \ell_n = \sum_{j \leq n, j \in D} (\sqrt{s_j} - \sqrt{s_{j-1}}).$$

Then $\sqrt{s_n} \leq \sqrt{a_n} + \ell_n$.

Proof. At an A -increment of size d , use $s_{j-1} \geq a_{j-1}$ and the fact that $\sqrt{x+d} - \sqrt{x}$ decreases in $x \geq 0$. Sum the resulting bound on the square-root increments in A and retain the exact increments in D . $\square$

Call a record short if it is initial or its preceding first-passage interval has width at most $N^{2/3}$. Define

$$T_{\text{sh}}(X) = \max \left(\{1\} \cup \left\{ \frac{M(N)^2}{N} : N \leq X, N \text{ a short record} \right\} \right), \quad b(X) = \sum_{h \leq X} \frac{2h-1}{h^2}.$$

Theorem 3.5 (Terminal-height recovery). *For $X \geq 1$,*

$$\sqrt{S(X)} \leq \sqrt{b(X)T_{\text{sh}}(X)} + 6, \quad (3.4)$$

$$S(X) \leq (4 \log X + 4)T_{\text{sh}}(X) + 72. \quad (3.5)$$

Proof. At a short record of height h , the ordinary envelope increment is at most $(M(N)^2/N)(2h - 1)/h^2$. Record heights are distinct and at most $\lfloor X \rfloor$, so their ordinary increments total at most $b(X)T_{\text{sh}}(X)$. Long records cost at most six in the square-root envelope by Theorem 3.3. Apply Lemma 3.4. Finally $b(X) = 2H_{\lfloor X \rfloor} - \sum_{h \leq X} h^{-2} \leq 2 \log X + 2$ and $(u + v)^2 \leq 2u^2 + 2v^2$ give the second inequality. $\square$

There is an adaptive version. Fix $0 < \beta < 1$ and $p \geq 1$; let $H(X) = (1 + S(X^\beta))^{p/2}$. Retain the initial record and every noninitial record with width $W < \sqrt{N}/H(N)$. Let T_{ret} be the envelope of $M(N)^2/N$ on the retained records, including one in its maximum.

Proposition 3.6 (Adaptive accounting). *The discarded records through X have total square-root-envelope increment at most $H(X) \log X$. Consequently*

$$\sqrt{S(X)} \leq \sqrt{b(X)T_{\text{ret}}(X)} + H(X) \log X. \quad (3.6)$$

Proof. At a discarded record, $W \geq \sqrt{N}/H(N)$; hence $N^{-1/2} \leq H(N)W/N \leq H(X) \int_m^N dt/t$. Disjointness of the first-passage intervals gives the claimed logarithmic budget. The retained ordinary increments are bounded exactly as in Theorem 3.5, and Lemma 3.4 finishes the proof. $\square$

These results allow arbitrary block lengths and arbitrarily long first-passage intervals. Their indexed objects are distinct: return blocks are excursions from a fixed level, whereas record intervals join successive first passages of absolute heights.

4 The field, source Hamiltonian, and horizon observable

The field and its exact horizon map are those of [6, Sections 1–10]. Let $\mathcal{E} = H^1(\mathbb{R})$ with inner product

$$\langle f, g \rangle_{\mathcal{E}} = \int_{\mathbb{R}} \left(\overline{f'(u)} g'(u) + \frac{1}{4} \overline{f(u)} g(u) \right) du. \quad (4.1)$$

The positive self-adjoint operator on $L^2(\mathbb{R})$ associated with this form is

$$H_0 = -\partial_u^2 + \frac{1}{4}, \quad \mathcal{D}(H_0) = H^2(\mathbb{R}).$$

Its Green function is $G_v(u) = e^{-|u-v|/2}$.

Proposition 4.1 (Reproducing field metric). *For $f \in \mathcal{E}$ and $v, w \in \mathbb{R}$,*

$$\langle G_v, f \rangle_{\mathcal{E}} = f(v), \quad \langle G_v, G_w \rangle_{\mathcal{E}} = e^{-|v-w|/2}, \quad \|G_v\|_{\mathcal{E}} = 1. \quad (4.2)$$

For arbitrary finite complex charges a_n ,

$$\phi_a(u) = \sum_{n \leq N} \frac{a_n}{\sqrt{n}} G_{\log n}(u) \implies \|\phi_a\|_{\mathcal{E}}^2 = \|a\|_G^2. \quad (4.3)$$

Proof. The derivative of G_v jumps by -1 at v , and it solves the homogeneous equation on either side. Hence $H_0 G_v = \delta_v$ in distributions. Integration by parts first for smooth compactly supported functions, followed by continuity in H^1 , proves the reproducing identity. For the last assertion use

$$\frac{e^{-|\log m - \log n|/2}}{\sqrt{mn}} = \frac{1}{\max(m, n)}$$

in the finite quadratic form. $\square$

In particular the actual arithmetic field satisfies

$$\phi_N = \sum_{n \leq N} \frac{\mu(n)}{\sqrt{n}} G_{\log n}, \quad H_0 \phi_N = J_N := \sum_{n \leq N} \frac{\mu(n)}{\sqrt{n}} \delta_{\log n}, \quad \|\phi_N\|_{\mathcal{E}}^2 = R(N). \quad (4.4)$$

Writing $M_N(x) = \sum_{n \leq \min(x, N)} \mu(n)$ and $h_N(x) = \sum_{n \leq \min(x, N)} \mu(n)/n$, with both zero below one, gives the two-channel formula

$$\phi_N(u) = F_N(u) + B_N(u), \quad F_N(u) = e^{-u/2} M_N(e^u), \quad B_N(u) = e^{u/2} (h(N) - h_N(e^u)). \quad (4.5)$$

Almost everywhere $\phi'_N = (B_N - F_N)/2$, so

$$|\phi'_N|^2 + \frac{1}{4} |\phi_N|^2 = \frac{1}{2} |F_N|^2 + \frac{1}{2} |B_N|^2. \quad (4.6)$$

Both square integrals on the right equal $R(N)$. The first is (2.2). For the second, expand $\int_0^\infty (h(N) - h_N(x))^2 dx$ and use $\min(m, n)/(mn) = 1/\max(m, n)$. Thus

$$R(N) = h(N)^2 + \sum_{k=1}^{N-1} (h(N) - h(k))^2. \quad (4.7)$$

The exterior fields are $h(N)e^{u/2}$ for $u < 0$ and $M(N)e^{-u/2}$ for $u > \log N$. Their field energies are $h(N)^2/2$ and $M(N)^2/(2N)$, respectively. All formulas refer to the completed field, including both exterior regions.

4.1 The Fourier readout

With $\hat{f}(k) = \int_{\mathbb{R}} f(u) e^{-iku} du$, $\hat{G}_0(k) = (k^2 + 1/4)^{-1}$. Plancherel therefore gives

$$R(N) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|D_N(\frac{1}{2} + ik)|^2}{k^2 + \frac{1}{4}} dk. \quad (4.8)$$

This can also be read entry by entry from

$$\frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{ikt}}{k^2 + 1/4} dk = e^{-|t|/2}.$$

Only finite source sums are interchanged with the integral.

The dilation generator in logarithmic coordinates is $H_{\text{dil}} = -i\partial_u$ on $H^1(\mathbb{R}) \subset L^2(\mathbb{R})$, and $H_0 = H_{\text{dil}}^2 + 1/4$. Its Fourier spectral parameter is real k and its field multiplier is $k^2 + 1/4$. These are the continuous spectral values of the free operator. The zeta zeros enter the arithmetic Dirichlet data and the final Mellin argument.

4.2 A normalized source and its prime occupations

Take $\mathcal{H} = \ell^2(\mathbb{N})$ with basis $|n\rangle$. Prime factorization identifies this basis with finite prime-occupation lists. The normalized source is

$$Z = \sum_{n \geq 1} \frac{\mu(n)^2}{n^2} = \prod_p (1 + p^{-2}) = \frac{15}{\pi^2}, \quad \Psi = \frac{1}{\sqrt{Z}} \sum_{n \geq 1} \frac{\mu(n)}{n} |n\rangle. \quad (4.9)$$

Equivalently, with the vacuum reference at each prime,

$$\Psi = \bigotimes_p \frac{|0\rangle_p - p^{-1}|1\rangle_p}{\sqrt{1 + p^{-2}}}. \quad (4.10)$$

The product converges because $\sum_p p^{-2} < \infty$. Its expansion has exactly the squarefree support and alternating multiplicative signs of μ ; higher prime occupations have zero amplitude.

Define the source Hamiltonian and its unitary group by

$$H_{\text{src}}|n\rangle = (\log n)|n\rangle, \quad \mathcal{D}(H_{\text{src}}) = \left\{ c : \sum_n (\log n)^2 |c_n|^2 < \infty \right\}, \quad U(\tau)c = (n^{-i\tau}c_n)_n. \quad (4.11)$$

The state belongs to the domain of every fixed power of this operator. Its norm and its Hamiltonian expectation are conserved; in particular $\langle \Psi, H_{\text{src}} \Psi \rangle = \sum_p \log p / (p^2 + 1) < \infty$.

4.3 The horizon map, adjoint, and exact modes

Define the finite-rank map $P_N : \mathcal{H} \rightarrow \mathcal{E}$ by

$$(P_N c)(u) = \sqrt{Z} \sum_{n \leq N} \sqrt{n} c_n G_{\log n}(u). \quad (4.12)$$

Then $P_N \Psi = \phi_N$. The normalizing weight in the map exactly compensates the normalization of the source.

Theorem 4.2 (Positive pullback and complete mode resolution). *The adjoint and pullback are*

$$(P_N^* f)_n = \sqrt{Zn} f(\log n) \mathbf{1}_{n \leq N}, \quad P_N^* P_N = Z \mathbf{B}_N \quad (4.13)$$

on the retained coordinates. The matrix $\mathbf{B}_N$ has the orthonormal eigenvectors and eigenvalues

$$\theta_{j,N} = \frac{(2j-1)\pi}{2N+1}, \quad v_{j,N}(n) = \frac{2}{\sqrt{2N+1}} \sin(n\theta_{j,N}), \quad \lambda_{j,N} = \frac{1}{4 \sin^2(\theta_{j,N}/2)}. \quad (4.14)$$

For $b_{j,N} = \sum_{n \leq N} \mu(n) v_{j,N}(n)/n$,

$$R(N) = \sum_{j=1}^N \lambda_{j,N} |b_{j,N}|^2. \quad (4.15)$$

Proof. The reproducing identity gives the adjoint. Its Gram entries are $Z \sqrt{mne}^{-|\log m - \log n|/2} = Z \min(m, n)$. Let C_N be the lower triangular matrix of ones. Since $\mathbf{B}_N = C_N C_N^T$, its inverse is tridiagonal with diagonal $2, \dots, 2, 1$ and adjacent entries -1 ; for $N = 1$ the inverse is (1) . The eigenproblem is the second-difference equation with boundary conditions $v_0 = 0$, $v_{N+1} = v_N$. Substituting $\sin(n\theta)$ gives eigenvalue $4 \sin^2(\theta/2)$ for the inverse and angles (4.14). The squared sine norm is $(2N+1)/4$, and distinct eigenvalues give orthogonality. Evaluation of the diagonalization on $c_n = \mu(n)/n$ proves (4.15). $\square$

The singular values of P_N are $\sqrt{Z \lambda_{j,N}}$. In particular $\|P_N\|^2 \sim 4ZN^2/\pi^2$. The conserved source norm and the arithmetic readout are linked by the cutoff-dependent positive observable $P_N^* P_N$.

Corollary 4.3 (Exact local inversion). *Let $w_n = \sqrt{n} \phi_N(\log n)$ for $1 \leq n \leq N$, and $w_0 = 0$. Then*

$$\mu(n) = n(2w_n - w_{n-1} - w_{n+1}) \quad (1 \leq n < N), \quad \mu(N) = N(w_N - w_{N-1}). \quad (4.16)$$

The identities hold for arbitrary finite charges in place of μ .

Proof. The sample vector is $w = \mathbf{B}_N c$, $c_n = \mu(n)/n$. Apply the tridiagonal inverse just computed. $\square$

4.4 Prime insertion with the inherited cutoff

Let $S_p|n\rangle = |pn\rangle$, and let $T_a\phi(u) = \phi(u - a)$. Directly on basis vectors,

$$P_N S_p = \sqrt{p} T_{\log p} P_{\lfloor N/p \rfloor}, \quad [H_{\text{src}}, S_p] = (\log p) S_p. \quad (4.17)$$

The commutator is understood on finitely supported source vectors.

For a finite prime set Q , let $\mu_Q(n)$ equal $\mu(n)$ when all prime factors of n belong to Q , and zero otherwise; include $\mu_Q(1) = 1$. Denote its field through N by $\phi_{Q,N}$.

Theorem 4.4 (Complete prime action). *If $p \notin Q$ and $K = \lfloor N/p \rfloor$, then*

$$\phi_{Q \cup \{p\}, N} = \phi_{Q, N} - p^{-1/2} T_{\log p} \phi_{Q, K}, \quad (4.18)$$

$$\|\phi_{Q \cup \{p\}, N}\|_{\mathcal{E}}^2 = \|\phi_{Q, N}\|_{\mathcal{E}}^2 + p^{-1} \|\phi_{Q, K}\|_{\mathcal{E}}^2 - 2p^{-1/2} \text{Re}\langle \phi_{Q, N}, T_{\log p} \phi_{Q, K} \rangle_{\mathcal{E}}. \quad (4.19)$$

Proof. Every squarefree product using p is uniquely pm , where m is a squarefree product from Q . Its sign is $-\mu_Q(m)$, weight $(pm)^{-1/2}$, and location $\log p + \log m$. The condition $pm \leq N$ is exactly $m \leq K$. Splitting the field sum gives (4.18). Expand its norm and use translation invariance. $\square$

On field families define $(\mathcal{D}_p \mathbf{f})_N = p^{-1/2} T_{\log p} f_{\lfloor N/p \rfloor}$, with $f_0 = 0$. These actions commute, since $\lfloor \lfloor N/p \rfloor / q \rfloor = \lfloor N/(pq) \rfloor$. For $g_N = G_0$ when $N \geq 1$ and $g_0 = 0$,

$$\phi_N = \left[\prod_{p \leq N} (I - \mathcal{D}_p) \mathbf{g} \right]_N. \quad (4.20)$$

This is the exact prime-color construction with all numerical horizons retained. The source also determines prime powers through

$$\Lambda(n) = \sum_{d|n} \mu(d) \log(n/d). \quad (4.21)$$

For a prime power, the two nonzero divisor contributions give $\log p$; for two or more distinct primes, expansion over their subsets cancels the constant and linear logarithmic terms. This proves the identity.

4.5 Phase transport

Let $R_N(\tau) = \|P_N U(\tau) \Psi\|_{\mathcal{E}}^2$.

Theorem 4.5 (Uniform phase comparison). *For $q(\tau) = \sqrt{1 + \tau^2} + |\tau|$,*

$$q(\tau)^{-2} R(N) \leq R_N(\tau) \leq q(\tau)^2 R(N). \quad (4.22)$$

The comparison holds for arbitrary finite complex charges.

Proof. In (4.8) the phase shifts the Dirichlet polynomial. Changing the integration variable gives

$$R_N(\tau) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|D_N(1/2 + it)|^2}{(t - \tau)^2 + 1/4} dt.$$

The ratio $(t^2 + 1/4)/((t - \tau)^2 + 1/4)$ is between $q(\tau)^{-2}$ and $q(\tau)^2$. These are the squared singular values of the shear $\begin{pmatrix} 1 & 2\tau \\ 0 & 1 \end{pmatrix}$ acting on $(t - \tau, 1/2)$. Integrate this pointwise comparison against the nonnegative spectral density. $\square$

For each fixed N , expansion of the finite phase sum also gives

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_N(\tau) d\tau = \sum_{n \leq N} \frac{\mu(n)^2}{n}. \quad (4.23)$$

All off-diagonal frequencies $\log(b/a)$ are nonzero and their long averages vanish. Equation (4.23) is a phase average at fixed N , with the original finite Dirichlet sum inside the readout.

5 Exact prefix carry and completed-prefix mode estimates

This section incorporates the signed prefix–tail comparison of [8, Sections 2–6]. Fix integers $1 \leq L, J \leq N$, put $c_n = \mu(n)/n$, and use the matrix B_N and modes of Theorem 4.2. Let V_N have columns $v_{j,N}$, and let Π_J retain the first J modal coordinates. Write

$$A = M(L), \quad I_L = I_M(L) = R(L) - A^2/L, \quad q = c^{\leq L} - (A/L)e_L, \quad w = c^{>L} + (A/L)e_L.$$

Theorem 5.1 (Exact boundary carry). *In the original energy metric,*

$$\|q\|_B^2 = I_L, \quad \langle q, w \rangle_B = 0, \quad R(N) = I_L + \|w\|_B^2. \quad (5.1)$$

In particular,

$$\|w\|_B^2 = \frac{A^2}{L} + 2A(h(N) - h(L)) + \|c^{>L}\|_B^2. \quad (5.2)$$

Proof. For every $b \geq L$, $\sum_{a \leq L} q_a \min(a, b) = \sum_{a \leq L} a q_a = 0$. Thus q is orthogonal to w . Its physical cumulative charge is $M(k)$ for $k < L$ and zero from L onward, giving its energy I_L . The original prefix–tail pairing factors through the total A , so expanding the remaining square proves (5.2). $\square$

The exterior field of the prefix equals $Ae^{-u/2}$ for $u \geq \log L$; the boundary charge in (5.2) has exactly that exterior field. The zero-charge residual retains the interior history. Its orthogonality in the full energy metric does not assert orthogonality after selecting finitely many modes.

5.1 A fifth-order factor for the historical remainder

Set

$$r = V_N^* q, \quad t = V_N^* c^{>L}, \quad d = V_N^* e_L/L, \quad b = V_N^* c = t + Ad + r.$$

For $k < L$ put

$$D_j(k) = \frac{v_{j,N}(k)}{k} - \frac{v_{j,N}(k+1)}{k+1}, \quad \Omega_{ij} = \sum_{k < L} k(k+1) D_i(k) D_j(k), \quad \Xi_{N,J}(L) = \text{tr}(\Omega_{ij})_{i,j \leq J}.$$

Abel summation, including the compensating endpoint, gives

$$r_j = \sum_{k < L} M(k) D_j(k), \quad \|\Pi_J r\|^2 \leq I_L \min\{\Xi_{N,J}(L), \lambda_{J,N}^{-1}\}. \quad (5.3)$$

The first estimate is Cauchy–Schwarz with historical weights; the second also uses $\sum_j \lambda_{j,N} |r_j|^2 = I_L$.

Theorem 5.2 (Historical projection factor). *For all N, J, L in the stated range,*

$$\Xi_{N,J}(L) \leq \frac{4}{2N+1} \frac{L^5 - 1}{45} \sum_{j \leq J} \theta_{j,N}^6, \quad (5.4)$$

$$\lambda_{J,N} \Xi_{N,J}(L) \leq \frac{\pi}{45} (L \theta_{J,N})^5. \quad (5.5)$$

Proof. For $g_j(x) = v_{j,N}(x)/x$ and $\kappa_N = 2/\sqrt{2N+1}$,

$$g'_j(x) = \kappa_N \frac{\theta_j x \cos(\theta_j x) - \sin(\theta_j x)}{x^2}.$$

Since $\sin z - z \cos z = \int_0^z t \sin t \, dt$, its absolute value is at most $z^3/3$. On each unit interval, weighted Cauchy–Schwarz gives $k(k+1)D_j(k)^2 \leq \int_k^{k+1} x^2 g'_j(x)^2 \, dx$. Integrate x^4 and sum to prove (5.4). Use $\lambda_J \theta_J^2 \leq \pi^2/4$, $\sum_{j \leq J} \theta_j^6 \leq J \theta_J^6$ and $\kappa_N^2 J L \leq 4L \theta_J / \pi$ for (5.5). $\square$

Let $\epsilon_{N,J,L}^2 = I_L \min\{\Xi_{N,J}(L), \lambda_{J,N}^{-1}\}$. The signed interaction and the complete population obey

$$\left| \operatorname{Re} \langle \Pi_J V_N^* c^{\leq L}, \Pi_J t \rangle - A \operatorname{Re} \langle \Pi_J d, \Pi_J t \rangle \right| \leq \epsilon_{N,J,L} \|\Pi_J t\|, \quad (5.6)$$

$$||\Pi_J b\| - \|\Pi_J(t + Ad)\|| \leq \epsilon_{N,J,L}. \quad (5.7)$$

These follow by substituting $V_N^* c^{\leq L} = Ad + r$ and applying Cauchy–Schwarz and the reverse triangle inequality. The expression $\|\Pi_J(t + Ad)\|^2$ retains the boundary square, the actual tail square, and their signed interaction together.

5.2 The optimal whole-prefix comparison

Define the $J \times J$ matrix

$$\mathcal{G}_{ij}(L) = \sum_{n=1}^L [v_{i,N}(n) - v_{i,N}(n-1)][v_{j,N}(n) - v_{j,N}(n-1)], \quad v_{j,N}(0) = 0.$$

Theorem 5.3 (Whole-prefix gain). *With $g_{N,J}(L) = \lambda_{J,N} \lambda_{\max}(\mathcal{G}(L))$,*

$$\lambda_{J,N} \|\Pi_J V_N^* c^{\leq L}\|^2 \leq g_{N,J}(L) R(L), \quad g_{N,J}(L) \leq \min \left\{ 1, \frac{4JL}{2N+1} \right\}. \quad (5.8)$$

The first constant is optimal for arbitrary coefficient vectors supported through L with prescribed energy. For the first mode it is

$$g_{N,1}(L) = \frac{2L + \sin(2L\theta_{1,N})/\sin\theta_{1,N}}{2N+1}. \quad (5.9)$$

Proof. The dual energy is $f^* \mathbf{B}_L^{-1} f = |f(1)|^2 + \sum_{n=2}^L |f(n) - f(n-1)|^2$. The Gram matrix of the first J functionals in this metric is $\mathcal{G}$, so its largest eigenvalue is the optimal squared norm. The full spectral resolution gives $g \leq 1$. Since

$$v_j(n) - v_j(n-1) = 2\kappa_N \sin(\theta_j/2) \cos((n - \frac{1}{2})\theta_j),$$

one has $\lambda_J \operatorname{tr} \mathcal{G} \leq \kappa_N^2 L \sum_{j \leq J} \lambda_J / \lambda_j \leq 4JL/(2N+1)$. For $J = 1$, summing the cosine squares yields (5.9). $\square$

5.3 A uniform growing-prefix compression

Put

$$L = \lfloor (N/J)^{5/6} \rfloor, \quad C_\partial = \frac{2\pi^6}{315}, \quad \mathcal{W}_{N,J} = \|\Pi_J(t + M(L)d)\|^2.$$

Theorem 5.4 (Population error at the natural scale). *For $1 \leq J \leq N$,*

$$\|\Pi_J r\|^2 \leq C_\partial \frac{J^2}{N^2}, \quad \left| \sqrt{\sum_{j \leq J} |b_{j,N}|^2} - \sqrt{\mathcal{W}_{N,J}} \right| \leq \sqrt{C_\partial} \frac{J}{N}. \quad (5.10)$$

The same bound holds for real physical charges of absolute value at most one, with their actual inherited total.

Proof. For such charges $I_L \leq L - H_L \leq L$. Convexity and midpoint integration give $\sum_{j=1}^J (2j-1)^6 \leq \int_0^J (2x)^6 dx = 64J^7/7$. Using (5.4),

$$I_L \Xi_{N,J}(L) \leq \frac{256\pi^6}{315} \frac{J^7 L^6}{(2N+1)^7} \leq \frac{2\pi^6}{315} \frac{J^7 L^6}{N^7} \leq C_\partial \frac{J^2}{N^2}.$$

The last step uses $L^6 \leq (N/J)^5$. Apply (5.7). This includes $L = 1$ and all integer rounding. $\square$

Equation (5.10) bounds the squared vector error. The retained population is a square containing the inherited charge; it is not replaced by a sum of separate unsigned allowances.

6 Orthogonal boundary compression and one chain allowance

We now use the full-metric projection of [9] and its shared-chain extension [11, Section 1]. Let $a_n \in \mathbb{R}$, $|a_n| \leq 1$, $c_n = a_n/n$, and $A(x) = \sum_{n \leq x} a_n$. For an integer partition $0 = x_0 < x_1 < \dots < x_m = N$, replace, on each cell $(a, b]$, the coefficient basis vector e_n by

$$T_{\mathcal{P}} e_n = \frac{b-n}{b-a} e_a + \frac{n-a}{b-a} e_b, \quad e_0 = 0. \quad (6.1)$$

The initial cell will be the singleton $(0, 1]$.

Proposition 6.1 (Two moments and orthogonal return). *Put $p = T_{\mathcal{P}} c$ and $z = c - p$. Then z is orthogonal in the B_N metric to every boundary source and hence to p . For a cell $(a, b]$ with $a > 0$, define $Q = \sum_{a < n \leq b} a_n$ and $H = \sum_{a < n \leq b} a_n/n$. Its endpoint contributions are*

$$p_a^{(a,b]} = \frac{bH - Q}{b-a}, \quad p_b^{(a,b]} = \frac{Q - aH}{b-a}. \quad (6.2)$$

They preserve both coefficient sum H and physical charge Q .

Proof. For a boundary $x_i \leq a$, the pairing of $e_n - \alpha_n e_a - \beta_n e_b$ with e_{x_i} is $x_i(1 - \alpha_n - \beta_n) = 0$. For a boundary $x_i \geq b$, it is $n - \alpha_n a - \beta_n b = 0$. There are no interior boundaries. Thus the residual is orthogonal to the boundary span, proving that $T_{\mathcal{P}}$ is its energy-orthogonal projection. Summing the two affine coefficients with weights c_n gives (6.2). Residuals from distinct cells are orthogonal because their potentials vanish outside their own cells. $\square$

Define the full potential

$$u_N(x) = \sum_{n \leq N} \frac{a_n}{n} \min(x, n), \quad x \geq 0. \quad (6.3)$$

The projection preserves its values at the boundaries and gives its linear interpolant between them. Its energy is

$$E_{\mathcal{P}}(N) = \|p\|_{\mathbf{B}}^2 = \sum_{r=1}^m \frac{[u_N(x_r) - u_N(x_{r-1})]^2}{x_r - x_{r-1}}. \quad (6.4)$$

Indeed the boundary Gram matrix factors as the cumulative-sum matrix, the diagonal matrix of interval lengths, and the transpose cumulative matrix. Its inverse gives exactly these squared differences.

Theorem 6.2 (A bounded full-source residual). *Set $d = N^{-1/6}$ and construct boundaries by*

$$x_0 = 0, \quad x_{r+1} = \min\{N, x_r + \max(1, \lfloor dx_r^{2/3} \rfloor)\}. \quad (6.5)$$

Then $m \leq 13\sqrt{N} + 2$ and

$$\|c\|_{\mathbf{B}}^2 = E_{\mathcal{P}}(N) + E_{\text{res}}(N), \quad 0 \leq E_{\text{res}}(N) \leq C_0 := 2^{-4/3}. \quad (6.6)$$

Moreover $E_{\mathcal{P}}(N) \geq A(N)^2/N$.

Proof. The residual kernel in a cell of length $h = b - a$ is the nonnegative bridge

$$G_{a,b}(n, m) = \min(n - a, m - a) - \frac{(n - a)(m - a)}{h}.$$

Endpoints contribute zero. For $a > 0$, $|c_n| \leq 1/a$, so finite summation gives the cell bound

$$0 \leq E_{(a,b)} \leq \frac{1}{a^2} \sum_{i,j=1}^{h-1} \left(\min(i, j) - \frac{ij}{h} \right) = \frac{h(h^2 - 1)}{12a^2} \leq \frac{h^3}{12a^2}.$$

On a nonsingleton mesh cell, $h \leq da^{2/3}$ and $b \leq 2a$; therefore

$$E_{(a,b)} \leq \frac{d^2}{12} ha^{-2/3} \leq \frac{2^{2/3}d^2}{12} \int_a^b t^{-2/3} dt.$$

Singleton cells have zero residual. Orthogonality and integration over the disjoint cells give $E_{\text{res}} \leq 2^{2/3}d^2N^{1/3}/4 = 2^{-4/3}$.

For the count, there are at most $\lceil (2/d)^{3/2} \rceil$ starts below $(2/d)^{3/2}$. Above it, every nontruncated interval has $h \geq (d/2)a^{2/3}$ and $b^{1/3} - a^{1/3} \geq d/(6 \cdot 2^{2/3})$. Sum the increments and allow one truncated final interval:

$$m \leq 6 \cdot 2^{2/3} \sqrt{N} + 2^{3/2} N^{1/4} + 2 \leq 13\sqrt{N} + 2.$$

Finally $u_N(0) = 0$, $u_N(N) = A(N)$, and Cauchy–Schwarz applied to (6.4) yields $E_{\mathcal{P}}(N) \geq A(N)^2/N$. $\square$

The inherited tail is visible in each slope:

$$\frac{u_N(b) - u_N(a)}{b - a} = \sum_{n>b} \frac{a_n}{n} + \frac{1}{b - a} \sum_{a < n \leq b} \frac{a_n(n - a)}{n}. \quad (6.7)$$

Both terms stay inside the same square in (6.4). Equivalently, E_{res} is the within-cell variance of u'_N in $\int_0^N |u'_N(x)|^2 dx$.

For the spectral coordinates of the same fixed projection, put $\mathcal{P}_{N,J} = \|\Pi_J V_N^* p\|^2$. Since $\lambda_{J,N} \geq N^2/(\pi^2 J^2)$, one has

$$\left| \sqrt{\sum_{j \leq J} |b_{j,N}|^2} - \sqrt{\mathcal{P}_{N,J}} \right| \leq \pi \sqrt{C_0} \frac{J}{N}. \quad (6.8)$$

This follows from the residual energy and the reverse triangle inequality. A single projected source serves every J .

6.1 One projection for all division-by-six moves

Fix $N_0 = N$ and $N_{j+1} = \lfloor N_j/6 \rfloor$, stopping at $N_r < 6$. Insert all these cutoffs into the mesh (6.5). Let $\Pi^{(N)}$ be the resulting orthogonal boundary projection and define

$$\widehat{R}^{(N)}(n) = \|\Pi^{(N)}c_n\|_{\mathbb{B}}^2, \quad \widehat{\delta}_j = \widehat{R}^{(N)}(N_j) - \widehat{R}^{(N)}(N_{j+1}), \quad \delta_j = R(N_j) - R(N_{j+1}).$$

Theorem 6.3 (Shared-chain error budget). *There are $\epsilon_j \geq 0$ such that*

$$\delta_j = \widehat{\delta}_j + \epsilon_j, \quad \sum_{j < r} \epsilon_j \leq C_0, \quad 0 \leq \sum_{j < r} [\delta_j]_+ - \sum_{j < r} [\widehat{\delta}_j]_+ \leq C_0. \quad (6.9)$$

The partition uses at most $13\sqrt{N} + 2 + r$ boundary locations.

Proof. Refining a boundary span cannot increase its orthogonal residual. Each chain cutoff is a boundary. Thus the residual of a prefix ending there is exactly the sum of the residuals of cells below that cutoff. Writing $e_j = R(N_j) - \widehat{R}^{(N)}(N_j)$, the difference $e_j - e_{j+1}$ is a sum of nonnegative cell energies in $(N_{j+1}, N_j]$. The initial mesh includes all integers through five, so $e_r = 0$. Hence $\epsilon_j = e_j - e_{j+1} \geq 0$ and their sum is $e_0 \leq C_0$. The positive-part assertion follows from $0 \leq [x + e]_+ - [x]_+ \leq e$ for $e \geq 0$. Each inserted cutoff adds at most one boundary. $\square$

There is also a complete formula for a retained edge $l = N_{j+1}$, $n = N_j$. Put $A = M(l)$, $h = h(n) - h(l)$, and, for $l \leq x \leq n$, $U_n(x) = M(x) + x \sum_{x < m \leq n} \mu(m)/m$. Then

$$\widehat{\delta}_j = 2Ah + lh^2 + \sum_{(a,b] \subset (l,n]} \frac{[U_n(b) - U_n(a)]^2}{b - a}. \quad (6.10)$$

The carried potential is linear on $[0, l]$, has value $A + lh$ at l , and equals U_n above l . Its interpolant therefore has energy $(A + lh)^2/l$ plus the displayed slope squares. Subtracting A^2/l proves (6.10). The projection preserves the boundary charge and the entire signed inherited term.

7 Completed multiplicative families and their signed complement

The results in this section are the collective and separated-family estimates of [10, Sections 2–3]. They connect the prime construction with the sixth-root scale used in Hypothesis 1.1. For weighted coefficient vectors define

$$\mathcal{A}_{d,N}e_m = d^{-1}e_{dm}\mathbf{1}_{dm \leq N}.$$

The cutoff is suppressed when the complete dilation fits. A complete dilation has energy

$$\|\mathcal{A}_d c\|_{\mathbb{B}}^2 = d^{-1}\|c\|_{\mathbb{B}}^2, \quad (7.1)$$

as follows directly from $\min(dm, dn) = d \min(m, n)$.

For a finite prime set C , put $q_C = \prod_{p \in C} p$ and define the coprime parent through K by

$$f_{C,K}(m) = \frac{\mu(m)}{m} \mathbf{1}_{m \leq K, (m, q_C) = 1}.$$

Lemma 7.1 (Exact recovery of the coprime parent). *For every $K \geq 1$,*

$$f_{C,K} = \sum_{\substack{d \leq K \\ p|d \Rightarrow p \in C}} \mathcal{A}_{d,K} c_K, \quad \|f_{C,K}\|_{\mathbb{B}} \leq \sqrt{S(K)} \prod_{p \in C} (1 - p^{-1/2})^{-1}. \quad (7.2)$$

Proof. The coefficient at n of the sum is $n^{-1} \sum_{d|n, p|d \Rightarrow p \in C} \mu(n/d)$. If a prime of C divides n , its local sum is zero; otherwise the sum is $\mu(n)$. This proves the identity. Its d th vector is a complete dilation of the source through $\lfloor K/d \rfloor$, so its squared norm is $R(\lfloor K/d \rfloor)/d \leq S(K)/d$. The triangle inequality and the finite-prime geometric product prove the norm estimate. $\square$

7.1 The complete two-prime comparison

The preceding identity permits a test of the entire 2, 3 multiplicative families, including all inherited numerical cutoffs. Write $f_K = f_{\{2,3\},K}$, set $f_0 = 0$, and define

$$S_6(X) = \max\{1, \|f_n\|_{\mathbb{B}}^2 : 1 \leq n \leq \lfloor X \rfloor\}.$$

This is the energy envelope of the actual coprime parents. The finite exceptional modifier (2.28) and the adaptive modifier (2.6) retain their earlier definitions.

Corollary 7.2 (Uniform comparison for the complete 2, 3 families). *For every $X \geq 1$,*

$$\frac{S_6(X)}{C_-^2} \leq S(X) \leq C_+^2 S_6(X), \quad (7.3)$$

$$C_+ = \left(1 + \frac{1}{\sqrt{2}}\right) \left(1 + \frac{1}{\sqrt{3}}\right), \quad C_- = \frac{1}{(1 - 1/\sqrt{2})(1 - 1/\sqrt{3})}.$$

In particular, $S(X) = X^{o(1)}$ if and only if $S_6(X) = X^{o(1)}$.

Proof. Unique factorization gives the finite identity

$$c_N = f_N - \mathcal{A}_2 f_{\lfloor N/2 \rfloor} - \mathcal{A}_3 f_{\lfloor N/3 \rfloor} + \mathcal{A}_6 f_{\lfloor N/6 \rfloor}. \quad (7.4)$$

Indeed every squarefree integer has a unique representation dm , where $d \mid 6$ and $(m, 6) = 1$, with sign $\mu(d)\mu(m)$; squareful integers have zero coefficient on both sides. Every dilation in (7.4) is complete at its displayed child cutoff. Equation (7.1) and the triangle inequality give $\sqrt{R(N)} \leq C_+ \sqrt{S_6(N)}$. Taking the envelope proves the upper comparison. Lemma 7.1 with $C = \{2, 3\}$ gives $\|f_N\|_{\mathbb{B}} \leq C_- \sqrt{S(N)}$ and hence the other comparison. Constants independent of X preserve every positive-exponent growth bound, proving the final assertion. $\square$

Thus the complete 2, 3 removal has a fixed norm cost in both directions. The finite neutralizer has the additive comparison (2.29), and bounded-width compression has (2.27). These facts transfer a proved growth estimate back to the original source. Their right sides still contain the actual parent or retained energy. The adaptive identity (2.14) identifies the historical energy that remains when endpoint masses cancel. No subpower upper estimate for that full quantity is inferred from endpoint cancellation alone.

7.2 Collective completed-family estimates

Proposition 7.3 (Collective completed-family estimate). *Let $K_C = \lfloor N/q_C \rfloor \geq 1$ and*

$$g_{C,N} = \prod_{p \in C} (I - \mathcal{A}_p) f_{C,K_C}, \quad \Lambda_C = \prod_{p \in C} \left(\frac{1 + p^{-1/2}}{1 - p^{-1/2}} \right)^2.$$

Every descendant fits through N , and

$$\|g_{C,N}\|_{\mathbb{B}}^2 \leq \Lambda_C S(K_C). \quad (7.5)$$

All parent overlaps are included in this bound.

Proof. The complete dilation identity gives $\|(I - \mathcal{A}_p)v\|_{\mathbf{B}} \leq (1 + p^{-1/2})\|v\|_{\mathbf{B}}$. Apply these inequalities to (7.2); the largest descendant is at most $q_C K_C \leq N$. $\square$

For arbitrary finite C , the cost is quantified by

$$\log \Lambda_C \leq 24\sqrt{\log(eq_C)}, \quad \Lambda_C \leq e^{\varepsilon+144/\varepsilon} q_C^\varepsilon \quad (\varepsilon > 0). \quad (7.6)$$

Indeed each logarithmic factor is at most $8/\sqrt{p}$. With $T = \log(eq_C) \geq e$, primes at most T contribute at most $2\sqrt{T}$ to $\sum p^{-1/2}$, and those above T at most $\log q_C / (\sqrt{T} \log T) \leq \sqrt{T}$. The finitely many smaller values $q_C = 1, 2, 3, 5$ satisfy the same bound directly. Finally $24\sqrt{T} \leq \varepsilon T + 144/\varepsilon$.

7.3 The full packet kernel

Define

$$P_C(y) = \sum_{d|q_C} \mu(d) \mathbf{1}_{d \leq y}, \quad K_C(m, n) = \sum_{d, e|q_C} \frac{\mu(d)\mu(e)}{\max(dm, en)}.$$

The kernel has the exact representations

$$\begin{aligned} K_C(m, n) &= \int_0^\infty P_C(x/m) P_C(x/n) \frac{dx}{x^2} \\ &= (mn)^{-1/2} k_C(\log(n/m)), \end{aligned} \quad (7.7)$$

$$\begin{aligned} k_C(u) &= \sum_{d, e|q_C} \frac{\mu(d)\mu(e)}{\sqrt{de}} e^{-|u + \log(e/d)|/2}, \\ \widehat{k}_C(\xi) &= \frac{\prod_{p \in C} |1 - p^{-1/2 + i\xi}|^2}{\xi^2 + 1/4}. \end{aligned} \quad (7.8)$$

The first formula expands the cumulative charge of a completed packet. The second follows by the logarithmic Green kernel, and Fourier transformation of each translated exponential proves the third. If C is nonempty, $K_C(m, n) = 0$ when $\max(m, n) \geq q_C \min(m, n)$, since the cumulative supports have disjoint interiors. Its diagonal is $K_C(m, m) = \kappa_C/m$, where $\kappa_C = \|\sum_{d|q_C} \mu(d)e_d\|_G^2$.

For the completed source and its actual complement $t = c_N - g_{C,N}$, the complete return is

$$R(N) = \|g_{C,N}\|_{\mathbf{B}}^2 + 2\langle g_{C,N}, t \rangle_{\mathbf{B}} + \|t\|_{\mathbf{B}}^2. \quad (7.9)$$

The packet kernel accounts for the internal term; the last two terms retain all unfinished and inherited interactions.

7.4 A completed family at the sixth-root scale

Lemma 7.4 (Separated zero-charge copies). *If h is supported through B and $\sum_n n h_n = 0$, then for every integer $p \geq B$,*

$$\langle h, \mathcal{A}_p h \rangle_{\mathbf{B}} = 0, \quad \|(I - \mathcal{A}_p)h\|_{\mathbf{B}}^2 = (1 + 1/p)\|h\|_{\mathbf{B}}^2. \quad (7.10)$$

Proof. Since $pm \geq n$ for $m, n \leq B$, the pairing is $p^{-1}(\sum_n n \overline{h_n})(\sum_m h_m) = 0$. Expand the square and use (7.1). $\square$

Theorem 7.5 (Sixth-root completed family). *For $N \geq 216$, choose the least prime $p > \lfloor N^{5/6}/6 \rfloor$, and put*

$$q = 6p, \quad K = \lfloor N/q \rfloor, \quad f(m) = \frac{\mu(m)}{m} \mathbf{1}_{m \leq K, (m,6)=1}, \quad h = (I - \mathcal{A}_2)(I - \mathcal{A}_3)f, \quad g = (I - \mathcal{A}_p)h.$$

Then $K \leq N^{1/6}$, every descendant is at most N , and

$$\|g\|_{\mathbb{B}}^2 = (1 + 1/p)\|h\|_{\mathbb{B}}^2 \leq A_* S(N^{1/6}), \quad A_* = \frac{18}{17}(17 + 12\sqrt{2})(7 + 4\sqrt{3}). \quad (7.11)$$

Writing $t = c_N - g$ gives the exact decomposition

$$R(N) = \|g\|_{\mathbb{B}}^2 + W_N, \quad W_N = 2\langle g, t \rangle_{\mathbb{B}} + \|t\|_{\mathbb{B}}^2. \quad (7.12)$$

The physical total of t is $M(N)$.

Proof. Bertrand's theorem gives $N^{5/6} < 6p \leq 2N^{5/6} \leq N$. Thus $K \leq N^{1/6}$. Moreover $p > N^{5/6}/6 \geq 6N^{1/6} \geq 6K$ and $p \geq 17$. The vector h has support through $6K$ and zero physical total, so Lemma 7.4 applies. Proposition 7.3 for $C = \{2, 3\}$ bounds its energy by $\Lambda_{23}S(K)$, where $\Lambda_{23} = (17 + 12\sqrt{2})(7 + 4\sqrt{3})$. This proves (7.11).

Since $p > K$, the parents are coprime to $6p$. Unique factorization shows that each selected squarefree output has exactly its actual Möbius coefficient; the other coefficients of g are zero. Every completed packet has physical total $\mu(m) \sum_{d|6p} \mu(d) = 0$, so the total of t is $M(N)$. Finally expand the full square to obtain (7.12). $\square$

The completed family also has the mode-population bound

$$\sum_{j \leq J} |\langle v_{j,N}, g \rangle|^2 \leq \frac{A_* S(N^{1/6})}{\lambda_{J,N}} \leq A_* \pi^2 \frac{J^2}{N^2} S(N^{1/6}). \quad (7.13)$$

This is the positive spectral resolution of its complete energy.

To display the signed complement in cumulative coordinates, put

$$V(x) = \sum_{\substack{m \leq K \\ (m,6)=1}} \mu(m) P_{\{2,3\}}(x/m), \quad B = 6K.$$

The cumulative charge of g is $V(x) - V(x/p)$. The two profiles have disjoint interiors because $p \geq B$. Consequently

$$\langle g, c_N \rangle_{\mathbb{B}} = \int_1^B \frac{V(x)}{x^2} \left(M(x) - \frac{M(px)}{p} \right) dx, \quad (7.14)$$

and the full expression is

$$\begin{aligned} W_N &= \int_1^N \frac{[M(x) - V(x) + V(x/p)]^2}{x^2} dx + \frac{M(N)^2}{N} \\ &\quad + 2 \int_1^{6K} \frac{V(x)}{x^2} \left[M(x) - \frac{M(px)}{p} - (1 + 1/p)V(x) \right] dx. \end{aligned} \quad (7.15)$$

These identities follow from the cumulative bilinear form and the substitution $x = py$ on the dilated support. $M(px)$ denotes its actual step-function value. Equation (7.15) keeps the complete inherited pairing and the unfinished terminal in one expression.

8 Positive circle domains and exact energy return

We use the circle construction of [12, Sections 1,3–4]. Let $a_1, \dots, a_N$ be real charges, and set $A = \sum a_n$, $h_a = \sum a_n/n$, $D_a(s) = \sum a_n n^{-s}$ and $R_a = \|\phi_a\|_{\mathcal{E}}^2$. Choose a prime $p > N$ and let $\ell = \log p$. On $L^2(0, \ell)$ define

$$\begin{aligned} H_{\ell, \theta} &= -\partial_u^2 + \frac{1}{4}, \\ \mathcal{D}(H_{\ell, \theta}) &= \{f \in H^2(0, \ell) : f(\ell) = e^{i\theta} f(0), \ f'(\ell) = e^{i\theta} f'(0)\}. \end{aligned} \quad (8.1)$$

Its form domain imposes the value condition. The associated energy of the point source at the logarithmic integers is

$$\mathcal{C}_{p, \theta}(a) = \frac{1}{\ell} \sum_{k \in \mathbb{Z}} \frac{|D_a(1/2 + i\kappa_k)|^2}{\kappa_k^2 + 1/4}, \quad \kappa_k = \frac{2\pi k + \theta}{\ell}. \quad (8.2)$$

Point masses act continuously on the form domain, and the series converges for every finite source. The source at the cut $u = 0$ is included once.

Theorem 8.1 (Two-phase isometry). *Put $s_p = \sqrt{p}$ and*

$$d_{p, \theta} = \operatorname{Re} \frac{s_p^{-1} e^{i\theta}}{1 - s_p^{-1} e^{i\theta}}, \quad w_0 = \frac{s_p - 1}{2s_p}, \quad w_\pi = \frac{s_p + 1}{2s_p}.$$

For every real finite source through N ,

$$\mathcal{C}_{p, \theta}(a) = R_a + 2d_{p, \theta} A h_a, \quad (8.3)$$

and

$$w_0 \mathcal{C}_{p, 0}(a) + w_\pi \mathcal{C}_{p, \pi}(a) = R_a. \quad (8.4)$$

The last equality holds at the kernel level for complex sources as well, and preserves every mixed inner product. For real sources,

$$\frac{s_p - \sqrt{N}}{s_p + 1} R_a \leq \mathcal{C}_{p, \theta}(a) \leq \frac{s_p + \sqrt{N}}{s_p - 1} R_a. \quad (8.5)$$

Thus $p \geq 4N$ gives $R_a/3 \leq \mathcal{C}_{p, \theta}(a) \leq 3R_a$. For Möbius charges and $p \geq (2N + 1)^2$,

$$|\mathcal{C}_{p, \theta}(\mu_{\leq N}) - R(N)| \leq 1. \quad (8.6)$$

Proof. The unitary endpoint relations annihilate the Green boundary form. Fourier series give the positive eigenvalues $\kappa_k^2 + 1/4$ and the absolutely convergent image kernel

$$K_{\ell, \theta}(d) = \sum_{j \in \mathbb{Z}} e^{-ij\theta} e^{-|d+j\ell|/2}.$$

All differences of source locations have magnitude below ℓ . For $j \geq 1$, the sign of $\log r - \log t + j\ell$ is fixed; after multiplication by $a_r a_t / \sqrt{rt}$ and summation its contribution is $p^{-j/2} e^{-ij\theta} h_a A$. Negative j give the conjugate. Sum the geometric series to obtain (8.3).

For the two phases, $d_{p, 0} = 1/(s_p - 1)$ and $d_{p, \pi} = -1/(s_p + 1)$. Since $w_0/(s_p - 1) - w_\pi/(s_p + 1) = 0$, their weighted kernel corrections cancel entry by entry. This proves (8.4) for all complex sources and hence for mixed inner products by polarization.

For the comparison, the line evaluation vectors $G_0, G_{\log N}$ have norm one and inner product $N^{-1/2}$; their readings are h_a and $A/\sqrt{N}$. The symmetric operator representing their product has eigenvalues $(N^{-1/2} \pm 1)/2$. Hence

$$\frac{1 - \sqrt{N}}{2} R_a \leq A h_a \leq \frac{1 + \sqrt{N}}{2} R_a.$$

Combine this with $-1/(s_p + 1) \leq d_{p,\theta} \leq 1/(s_p - 1)$ to obtain (8.5). Its coefficients are bounded by $1/3$ and 3 when $s_p \geq 2\sqrt{N}$.

Finally the divisor identity gives

$$1 = \sum_{n \leq N} \mu(n) \lfloor N/n \rfloor = N h(N) - \sum_{n \leq N} \mu(n) \{N/n\}.$$

The $n = 1$ fractional part is zero; bounding the other $N - 1$ terms proves $|h(N)| \leq 1$. With $|M(N)| \leq N$, the correction is at most $2N/(s_p - 1) \leq 1$ under the stated condition. $\square$

The direct sum of the two circle solution spaces, with factors $\sqrt{w_0}$ and $\sqrt{w_\pi}$, is therefore an isometric image of the finite arithmetic field space. When comparing two cutoffs, the same prime and the same weights are used for both.

Corollary 8.2 (One circle for a full chain). *Fix N , choose one prime $p \geq (2N + 1)^2$, and use it throughout the chain of Theorem 6.3. Let $\tilde{\delta}_j = \mathcal{C}_{p,\pi}(\mu_{\leq N_j}) - \mathcal{C}_{p,\pi}(\mu_{\leq N_{j+1}})$. Then*

$$\sum_{j < r} |\tilde{\delta}_j - \delta_j| \leq \frac{7}{5}, \quad \left| \sum_{j < r} [\tilde{\delta}_j]_+ - \sum_{j < r} [\delta_j]_+ \right| \leq \frac{7}{5}. \quad (8.7)$$

Combining with the shared boundary projection costs at most $7/5 + C_0$ in total. The two-phase readout has zero return error.

Proof. At a child cutoff n the antiperiodic error is $e_n = -2M(n)h(n)/(s_p + 1)$, with $|e_n| \leq 2n/(s_p + 1)$. Its variation along the chain is bounded by

$$\frac{2}{s_p + 1} \left(N + 2 \sum_{j \geq 1} N/6^j \right) = \frac{2N}{s_p + 1} \frac{7}{5} \leq \frac{7}{5}.$$

The positive-part function is 1-Lipschitz. The last assertion uses (6.9) and the triangle inequality once over the whole chain. $\square$

8.1 The scaling potential and its positive domain

For Möbius charges let

$$V_N(x) = \sum_{n \leq N} \frac{\mu(n)}{n} \min(x, n), \quad f_N^{\text{sc}}(u) = V_N(e^u) = e^{u/2} \phi_N(u), \quad \Delta_{\text{sc}} = \partial_u^2 - \partial_u.$$

The derivative of V_N jumps by $-\mu(n)/n$ at n , and the derivative of f_N^{sc} jumps by $-\mu(n)$ at $\log n$. Away from these locations the scaling differential expression vanishes. Thus, distributionally,

$$-\Delta_{\text{sc}} f_N^{\text{sc}} = \sum_{n \leq N} \mu(n) \delta_{\log n}, \quad e^{-u/2} (-\Delta_{\text{sc}}) e^{u/2} = H_0. \quad (8.8)$$

For $f = e^{u/2}\psi$, integration of the cross term gives

$$\int_a^b (|\psi'|^2 + \frac{1}{4}|\psi|^2) du = \int_a^b e^{-u}|f'|^2 du - \frac{1}{2}[e^{-u}|f|^2]_a^b. \quad (8.9)$$

For the arithmetic field both full-line boundary terms vanish, and this recovers (4.7).

On the circle, the positive weighted domain is specified by

$$f(u + \ell) = \sqrt{p} e^{i\theta} f(u), \quad f'(u + \ell) = \sqrt{p} e^{i\theta} f'(u). \quad (8.10)$$

Multiplication by $e^{-u/2}$ is unitary from $L^2((0, \ell), e^{-u} du)$ to $L^2(0, \ell)$ and sends this domain to (8.1). The boundary term in (8.9) vanishes, so the positive form is preserved. The operator domain and its density weight are part of the construction.

8.2 A fixed circle with every winding character

Theorem 8.3 (Faithful all-character return). *For any fixed $\ell > 0$, define on finite arithmetic fields*

$$(\mathcal{Z}_\ell \phi)(t, \theta) = \sum_{j \in \mathbb{Z}} e^{-ij\theta} \phi(t + j\ell), \quad 0 \leq t \leq \ell, \quad 0 \leq \theta < 2\pi.$$

It extends to an isometry from $\mathcal{E}$ onto the direct integral of the circle form spaces with transition $e^{i\theta}$, with

$$\|\phi\|_{\mathcal{E}}^2 = \frac{1}{2\pi} \int_0^{2\pi} \int_0^\ell (|\partial_t \mathcal{Z}_\ell \phi|^2 + \frac{1}{4}|\mathcal{Z}_\ell \phi|^2) dt d\theta. \quad (8.11)$$

The inverse reads the Fourier coefficient of each winding interval.

Proof. For smooth compactly supported fields the sum is finite. Parseval in the winding index proves (8.11) separately for the field and its derivative. At the cut the value satisfies $F(\ell, \theta) = e^{i\theta} F(0, \theta)$. Conversely the Fourier coefficients in θ reconstruct the intervals and this relation matches their H^1 traces. Density gives the Hilbert-space extension. Finite arithmetic fields have exponential tails, so their displayed series converge directly. $\square$

All prime translations are preserved. For $T_a \phi(u) = \phi(u - a)$, let $k = \lfloor (t - a)/\ell \rfloor$ and $r = t - a - k\ell \in [0, \ell)$. Then

$$(\mathcal{T}_a F)(t, \theta) = e^{ik\theta} F(r, \theta), \quad \mathcal{Z}_\ell T_a = \mathcal{T}_a \mathcal{Z}_\ell.$$

These operators are unitary and satisfy $\mathcal{T}_a \mathcal{T}_b = \mathcal{T}_{a+b}$. Thus the prime action (4.18), including its child cutoff and its mixed inner product, passes through the fixed-circle map unchanged.

9 The complete derivative record in a fixed Hardy space

This section assembles the exact infinity- and quadratic-plane construction of [14, Sections 3–10]. The disk is fixed for every arithmetic cutoff. Set

$$s = \frac{1}{1 - z}, \quad f_N(z) = D_N \left(\frac{1}{1 - z} \right) = \sum_{k=0}^{\infty} \gamma_{k,N} z^k, \quad |z| < 1. \quad (9.1)$$

The map sends $\mathbb{D}$ to $\operatorname{Re} s > 1/2$ and sends its center to $s = 1$.

Theorem 9.1 (Exact Hardy energy). *For every finite arithmetic source,*

$$R(N) = \|f_N\|_{H^2(\mathbb{D})}^2 = \sum_{k \geq 0} |\gamma_{k,N}|^2 = \text{tr } \mathcal{Q}_N, \quad \mathcal{Q}_N = |\gamma_N\rangle\langle\gamma_N|. \quad (9.2)$$

The corresponding map preserves the mixed inner products of arbitrary finite complex sources.

Proof. On $z = e^{i\theta}$,

$$s = \frac{1}{2} + \frac{i}{2} \cot \frac{\theta}{2}, \quad |d\theta| = \frac{|dt|}{t^2 + 1/4}, \quad t = \frac{1}{2} \cot(\theta/2).$$

For finite N , f_N is bounded analytic because $|n^{-1/(1-z)}| \leq n^{-1/2}$. Its radial boundary integral is therefore legitimate. The change of variable converts its normalized boundary square integral exactly to (4.8). Parseval for its Taylor series proves the coefficient identity. The rank-one positive operator has trace equal to the squared coefficient norm. Apply the same finite kernel calculation to mixed inner products. $\square$

9.1 Derivative and quadratic coordinates

The original complete jet of the entire Dirichlet polynomial at 1 is

$$\eta_{j,N} = \frac{D_N^{(j)}(1)}{j!} = \frac{(-1)^j}{j!} \sum_{n \leq N} \frac{\mu(n)(\log n)^j}{n}. \quad (9.3)$$

Its exact row-finite transformation to Hardy coefficients is

$$\gamma_{0,N} = \eta_{0,N}, \quad \gamma_{k,N} = \sum_{j=1}^k \binom{k-1}{j-1} \eta_{j,N} \quad (k \geq 1). \quad (9.4)$$

Indeed $s-1 = z/(1-z)$ and $(z/(1-z))^j = \sum_{k \geq j} \binom{k-1}{j-1} z^k$. The triangular transformation is invertible on formal power series. Every output row uses finitely many derivatives.

The full quadratic plane consists of all mixed products $\gamma_{k,N} \overline{\gamma_{l,N}}$. For example,

$$(\mathcal{Q}_N)_{kk} = \sum_{i,j=1}^k \binom{k-1}{i-1} \binom{k-1}{j-1} \eta_{i,N} \overline{\eta_{j,N}} \quad (k \geq 1).$$

The finite row is assembled before summing its energy over k . The resulting sum converges by Theorem 9.1. A nonconstant polynomial truncation in $s-1$ would instead produce a pole at $z=1$ and a nonintegrable boundary square. The operation used here transports the complete function before selecting Hardy coordinates.

An explicit polynomial form is

$$P_0(x) = 1, \quad P_k(x) = \sum_{j=1}^k \binom{k-1}{j-1} \frac{(-x)^j}{j!}, \quad \gamma_{k,N} = \sum_{n \leq N} \frac{\mu(n)}{n} P_k(\log n). \quad (9.5)$$

Their generating function is

$$\sum_{k \geq 0} P_k(x) z^k = \exp\left(-\frac{xz}{1-z}\right). \quad (9.6)$$

These are $L_k^{(-1)}$, in the usual generalized Laguerre notation [1, Eq. 18.12.13]. The mixed Hardy identity gives the complete kernel relation

$$\sum_{k \geq 0} P_k(x) P_k(y) = e^{\min(x,y)} \quad (x, y \geq 0). \quad (9.7)$$

To see this, apply the boundary integral to e^{-xs} and e^{-ys} ; their mixed norm is $e^{-\max(x,y)}$. Multiplication by e^{x+y} gives (9.7). Cauchy–Schwarz controls the mixed coefficient series.

9.2 Finite and growing derivative populations

Define

$$\mathcal{H}_K(N) = \sum_{k=0}^K |\gamma_{k,N}|^2, \quad \mathcal{T}_K(N) = \sum_{k>K} |\gamma_{k,N}|^2, \quad R(N) = \mathcal{H}_K(N) + \mathcal{T}_K(N). \quad (9.8)$$

Proposition 9.2 (Finite-jet allowance). *For every fixed $K \geq 0$,*

$$\mathcal{H}_K(N) \leq 1 + 4 \sum_{k=1}^K \left[\sum_{j=1}^k \binom{k-1}{j-1} \frac{(\log N)^j}{j!} \right]^2 \leq C_K (1 + \log N)^{2K}, \quad (9.9)$$

where C_K is independent of N .

Proof. The divisor argument in Theorem 8.1 gives $|h(N)| \leq 1$. For $j \geq 1$, partial summation yields

$$\sum_{n \leq N} \frac{\mu(n)(\log n)^j}{n} = h(N)(\log N)^j - j \int_1^N h(x) \frac{(\log x)^{j-1}}{x} dx.$$

Thus $|\eta_{0,N}| \leq 1$ and $|\eta_{j,N}| \leq 2(\log N)^j/j!$. Insert these bounds in (9.4) and sum the squares. One may take $C_K = 1 + 4 \sum_{k=1}^K (\sum_{j=1}^k \binom{k-1}{j-1}/j!)^2$. $\square$

Proposition 9.3 (Growing-head allowance). *For $N \geq 3$ and $K \geq 1$, with $x = \log N$,*

$$\mathcal{H}_K(N) \leq 1 + 4Kx^2 e^{4\sqrt{Kx}}. \quad (9.10)$$

Consequently, for $0 < \gamma < 1$ and $K_N = \lfloor (\log N)^\gamma \rfloor$,

$$\mathcal{H}_{K_N}(N) \leq 1 + 4(\log N)^{\gamma+2} \exp\{4(\log N)^{(1+\gamma)/2}\} = N^{o(1)}. \quad (9.11)$$

Proof. The inner sum in (9.9) is at most

$$x \sum_{m=0}^{k-1} \frac{(kx)^m}{m!(m+1)!} \leq x e^{2\sqrt{kx}}.$$

The first inequality uses $\binom{k-1}{j-1} \leq k^{j-1}/(j-1)!$; the second bounds this sum by the diagonal terms in $e^{\sqrt{kx}} e^{\sqrt{kx}}$. Square and sum over $k \leq K$. The final conclusion follows because $(1+\gamma)/2 < 1$. $\square$

At a scale step use $L = L_N$ and the same degree K at both cutoffs. Then

$$\delta_N = \mathcal{H}_K(N) - \mathcal{H}_K(L) + \mathcal{T}_K(N) - \mathcal{T}_K(L), \quad (9.12)$$

$$\begin{aligned} [\delta_N]_+ &\leq \mathcal{H}_K(N) + [\mathcal{T}_K(N) - \mathcal{T}_K(L)]_+, \\ [\mathcal{T}_K(N) - \mathcal{T}_K(L)]_+ &\leq [\delta_N]_+ + \mathcal{H}_K(L). \end{aligned} \quad (9.13)$$

The signed difference of the full quadratic traces is exactly δ_N . Its tail expression is

$$\sum_{k>K} \left[\left| \sum_{n \leq N} \frac{\mu(n)}{n} P_k(\log n) \right|^2 - \left| \sum_{n \leq L} \frac{\mu(n)}{n} P_k(\log n) \right|^2 \right]. \quad (9.14)$$

Both square sums converge, so their difference is absolutely summable as a sequence of differences. The head allowances are proved in (9.9)–(9.11); a bound for the tail increment is a corresponding formulation of the growth premise.

9.3 Prime insertions in the disk

For $a \geq 0$ let

$$I_a(z) = \exp \left[-\frac{a}{2} \frac{1+z}{1-z} \right]. \quad (9.15)$$

It is bounded by one in the disk and has boundary modulus one almost everywhere. Thus its multiplication operator is an isometry on H^2 , and $I_a I_b = I_{a+b}$, $n^{-1/(1-z)} = n^{-1/2} I_{\log n}(z)$. The complete arithmetic insertion becomes

$$f_{Q \cup \{p\}, N} = f_{Q, N} - p^{-1/2} I_{\log p} f_{Q, \lfloor N/p \rfloor}. \quad (9.16)$$

It is (4.18) in the fixed disk metric, including the child horizon. In particular the relative phases and the full quadratic interactions of all colors are retained.

The disk boundary in this section compactifies the spectral line. The circles in Section 8 periodize the spatial logarithmic coordinate. Their common arithmetic norm is precisely (9.2) and (8.4); no identification of their different coordinates is required.

10 A calibrated infinite derivative lift and its quadratic readout

We also retain the uniformly conditioned field-valued lift of [15, Sections 2–5]. Write $k_n(u) = n^{-1/2} G_{\log n}(u)$ and $\mathcal{E}_N = \text{span}\{k_1, \dots, k_N\} \subset \mathcal{E}$. For $N \geq 2$ define the normalized logarithmic generator by

$$\mathcal{L}_N \left(\sum_{n \leq N} a_n k_n \right) = \sum_{n \leq N} \frac{\log n}{\log N} a_n k_n. \quad (10.1)$$

All norms and adjoints below use the original Green metric.

Theorem 10.1 (Generator bound). *For every $N \geq 2$,*

$$\|\mathcal{L}_N\|_{\mathcal{E}_N \rightarrow \mathcal{E}_N} \leq 1 + \frac{2}{\log N}. \quad (10.2)$$

Proof. Let $A(x) = \sum_{n \leq \min(x, N)} a_n$ and $w(x) = \min(\log x / \log N, 1)$ for $x \geq 1$. Abel summation gives the cumulative weighted charge

$$A_w(x) = w(x)A(x) - \frac{1}{\log N} \int_1^{\min(x, N)} \frac{A(t)}{t} dt.$$

Multiplication by w has norm at most one in $L^2(dx/x^2)$. For $f(t) = A(t)t^{-1}\mathbf{1}_{[1, N]}(t)$, extended by zero below one, Hardy's inequality is

$$\int_0^\infty \left| \frac{1}{x} \int_0^x f(t) dt \right|^2 dx \leq 4 \int_0^\infty |f(t)|^2 dt.$$

Writing $F(x) = \int_0^x f$, integration by parts yields $\int |F|^2/x^2 = 2 \operatorname{Re} \int \bar{F} f/x$; the boundary terms vanish for this compactly supported f , and Cauchy–Schwarz gives the displayed inequality. The cumulative norm identity and the triangle inequality now prove (10.2). $\square$

For $N \geq 6$ fix

$$d_* = 1 + \frac{2}{\log 6}, \quad T_N = \frac{\mathcal{L}_N}{2d_*}, \quad \lambda_N^{\text{jet}} = 2d_* \log N. \quad (10.3)$$

Then $\|T_N\| \leq 1/2$. The entire field-valued orbit is

$$\Phi_{a, N}(z) = e^{zT_N} \phi_a = \sum_{n \leq N} a_n n^{z/\lambda_N^{\text{jet}}} k_n, \quad C_j(a; N) = \frac{T_N^j \phi_a}{j!}.$$

Theorem 10.2 (Complete derivative norm and calibration). *The map*

$$V_N \phi = (T_N^j \phi / j!)_{j \geq 0} \in \ell^2(\mathbb{N}_0; \mathcal{E}_N)$$

satisfies

$$\|\phi\|^2 \leq \|V_N \phi\|^2 \leq \kappa \|\phi\|^2 < \frac{4}{3} \|\phi\|^2 \quad (\phi \neq 0), \quad \kappa = \sum_{j=0}^{\infty} \frac{4^{-j}}{(j!)^2}. \quad (10.4)$$

For every $r > 0$, $\sum_j \|C_j\| r^j \leq e^{r/2} \|\phi\|$. If

$$\mathcal{A}_N = V_N^* V_N, \quad \mathcal{U}_N = V_N \mathcal{A}_N^{-1/2}, \quad (10.5)$$

then $\mathcal{U}_N^ \mathcal{U}_N = I$. The inverse on its image is $\phi = \mathcal{A}_N^{1/2} \pi_0 \mathcal{U}_N \phi$, where π_0 reads the zeroth derivative coordinate.*

Proof. The zeroth coordinate equals ϕ , while $\|T_N^j \phi\| \leq 2^{-j} \|\phi\|$. Sum its squared norm and its weighted norm series. Since $\kappa < \sum_{j \geq 0} 4^{-j} = 4/3$, the constants follow. The operator series $\mathcal{A}_N = \sum_{j \geq 0} (T_N^*)^j T_N^j / (j!)^2$ converges in norm and satisfies $I \leq \mathcal{A}_N \leq \kappa I$. Functional calculus proves the isometry. The formula for the inverse follows because $\pi_0 V_N = I$. $\square$

The uncalibrated quadratic lift is the positive trace-class operator $|V_N \phi\rangle \langle V_N \phi|$. Its (j, l) block is $|C_j\rangle \langle C_l|$, so all derivative cross terms are present. After calibration,

$$\langle \mathcal{U}_N \phi, \mathcal{U}_N \psi \rangle = \langle \phi, \psi \rangle_{\mathcal{E}}. \quad (10.6)$$

The Taylor translation law is convergent:

$$C_n(z+h) = \sum_{k \geq 0} \binom{n+k}{n} C_{n+k}(z) h^k, \quad \|e^{hT_N}\| \leq e^{|h|/2}.$$

An unnormalized shift $a_n \mapsto n^w a_n$ corresponds to $h = \lambda_N^{\text{jet}} w$; this scale conversion remains explicit.

All lifted spaces sit inside $\ell^2(\mathbb{N}_0; \mathcal{E})$. When a prefix and tail are compared, both use the same ambient N , generator, calibration, and circle map. The two-phase circle isometry may then be applied to every derivative coordinate without changing (10.6).

10.1 The complete scale coefficient as a signed trace

For $L = L_N$ set $v = \phi_L$, $t = \phi_N - \phi_L$ and $b = \mathcal{B}(N^{1/6})$. Their full increment is

$$\delta_N = \|t\|_{\mathcal{E}}^2 + 2 \operatorname{Re} \langle v, t \rangle_{\mathcal{E}} = \|t\|_{\mathcal{E}}^2 + 2M(L)(h(N) - h(L)). \quad (10.7)$$

Define in the calibrated joint space

$$a = \frac{\mathcal{U}_N v}{b^{5/2}}, \quad d = \frac{\mathcal{U}_N t}{b^{5/2}}, \quad \mathfrak{D}_N = |a + d\rangle \langle a + d| - |a\rangle \langle a|. \quad (10.8)$$

Then

$$\operatorname{tr} \mathfrak{D}_N = \frac{\delta_N}{\mathcal{B}(N^{1/6})^5}, \quad \mathfrak{c}_N = [\operatorname{tr} \mathfrak{D}_N]_+. \quad (10.9)$$

The trace is taken before its positive part.

Theorem 10.3 (Two-mode signed readout). *The operator $\mathfrak{D}_N$ is self-adjoint and has rank at most two. Put*

$$p_0 = \|a\|^2, \quad q_0 = \|d\|^2, \quad z_0 = \langle a, d \rangle, \quad \Delta = q_0 + 2 \operatorname{Re} z_0, \quad \omega = p_0 q_0 - |z_0|^2 \geq 0.$$

Its possible nonzero eigenvalues are

$$\nu_{\pm} = \frac{\Delta \pm \sqrt{\Delta^2 + 4\omega}}{2}, \quad \mathfrak{c}_N = [\nu_+ + \nu_-]_+. \quad (10.10)$$

Proof. The range lies in $\operatorname{span}\{a, d\}$. If $a \neq 0$, decompose d into its component along $a/\|a\|$ and one orthogonal component. The resulting Hermitian two-by-two matrix has trace Δ and determinant $-\omega$, giving (10.10). The cases $a = 0$ and linearly dependent vectors follow directly, allowing zero eigenvalues. $\square$

The original horizon observable also passes through this lift exactly:

$$(\mathcal{U}_N P_N)^* (\mathcal{U}_N P_N) = P_N^* P_N = Z \mathcal{B}_N. \quad (10.11)$$

Thus the growth premise of Theorem 1.2 has the precise joint-plane form

$$\operatorname{tr} \mathfrak{D}_N \leq C(1 + \log N)^a e^{A(\log N)^\rho}. \quad (10.12)$$

The transfer cost in (10.4) is a uniform comparison of representations. Equation (10.12) concerns the actual signed arithmetic trace preserved by them.

11 Finite descent and the Mellin conclusion

This section assembles the closing proofs of [13, Section 3 and Section 7] and [14, Section 8]. The only arithmetic allowance used is Hypothesis 1.1.

Lemma 11.1 (Logarithmic descent). *Under Hypothesis 1.1, the bounds (1.7) hold.*

Proof. Starting at an integer $n \geq 6$, repeatedly replace it by $\lfloor n/6 \rfloor$ until reaching the finite base range. There are at most $\log n / \log 6$ increments. Since every factor in the right side of (1.6) is nondecreasing with its cutoff, summing and taking the envelope gives, for a fixed C_1 ,

$$\mathcal{B}(X) \leq C_1(1 + \log X)^{a+1} e^{A(\log X)^\rho} \mathcal{B}(X^{1/6})^5. \quad (11.1)$$

The constant includes the finite base range. Put $F(t) = \log \mathcal{B}(e^t)$. For $t \geq 1$,

$$F(t) \leq 5F(t/6) + C_2 + (a+1)\log(1+t) + At^\rho. \quad (11.2)$$

Let $r = \lceil \log_6 t \rceil$ and stop after these finitely many iterations. The base term is $O(5^r) = O(t^\alpha)$. For the logarithmic term, put $u = t/6^r \in (1/6, 1]$ and change index to $k = r - j$. Then

$$\sum_{j < r} 5^j \log(1 + t/6^j) \leq 5^r \sum_{k \geq 1} 5^{-k} (\log 2 + k \log 6) = O(t^\alpha).$$

The constant term has the same order. The additional term is

$$At^\rho \sum_{j=0}^{r-1} (5/6^\rho)^j.$$

This is $O(t^\alpha)$ when $\rho < \alpha$, $O(t^\alpha \log t)$ when $\rho = \alpha$, and $O(t^\rho)$ when $\rho > \alpha$. This proves (1.7); if $A = 0$ the extra term is absent. $\square$

11.1 An explicit constant-coefficient version

Theorem 11.2 (Fixed coefficient). *Suppose $\delta_N \leq c_* \mathcal{B}(N^{1/6})^5$ for all $N \geq 6$, where $c_* \geq 0$ is fixed. Define*

$$C_* = \frac{73}{30} + \frac{c_*}{\log 6}, \quad K(c_*) = 5 \left(\frac{5}{4} \log 2 + \frac{1}{4} \log C_* + \frac{5}{16} \log 6 \right). \quad (11.3)$$

Then for $X \geq e$,

$$\mathcal{B}(X) \leq \exp\{K(c_*)(\log X)^\alpha\}. \quad (11.4)$$

If instead the coefficient is $c_(1 + \log N)^a$, one may take*

$$K_a = 5 \left[\log 2 + \frac{1}{4} \log C_* + (a+1) \left(\frac{1}{4} \log 2 + \frac{5}{16} \log 6 \right) \right]. \quad (11.5)$$

Proof. The retained exact base value is $S(5) = 43/30$ [13, Theorem 3.1]. The division-by-six telescope gives $R(n) \leq 43/30 + (c_* \log n / \log 6) \mathcal{B}(n^{1/6})^5$ and therefore $\mathcal{B}(X) \leq C_*(1 + \log X) \mathcal{B}(X^{1/6})^5$. For $0 \leq t \leq 1$, $F(t) = \log 2$. Iterating the logarithmic recurrence as in Lemma 11.1 and using

$$\sum_{k \geq 1} 5^{-k} = \frac{1}{4}, \quad \sum_{k \geq 1} k 5^{-k} = \frac{5}{16}, \quad 5^r \leq 5t^\alpha$$

gives (11.3)–(11.4). The polylogarithmic hypothesis replaces $\log(1+t)$ by $(a+1)\log(1+t)$ in the same argument, yielding (11.5). $\square$

The exponent α comes from the pair $(5, 6)$ in the scale recurrence. In logarithmic coordinate $u = \log N$, its smooth comparison allowance $\mathcal{A}(u) = e^{Ku^\alpha}$ satisfies

$$\frac{d}{du} \log \mathcal{A}(u) = K\alpha u^{\alpha-1}, \quad \frac{d}{d(u^\alpha)} \log \mathcal{A}(u) = K. \quad (11.6)$$

This is the constant rate in the subpower coordinate supplied by the existing descent. It is a consequence of the scale allowance.

11.2 Transfer of the hypothesis through the representations

The exact identities collected above give, with one common prime $p > N$ for the circle readings,

$$\begin{aligned} \delta_N &= 2M(L_N)(h(N) - h(L_N)) + \|\phi_N - \phi_{L_N}\|_{\mathcal{E}}^2 \\ &= \text{tr}(\mathcal{Q}_N - \mathcal{Q}_{L_N}) = \mathcal{B}(N^{1/6})^5 \text{tr} \mathfrak{D}_N \\ &= w_0[\mathcal{C}_{p,0}(\mu_{\leq N}) - \mathcal{C}_{p,0}(\mu_{\leq L_N})] + w_\pi[\mathcal{C}_{p,\pi}(\mu_{\leq N}) - \mathcal{C}_{p,\pi}(\mu_{\leq L_N})]. \end{aligned} \quad (11.7)$$

Thus the same upper allowance on any complete expression in (11.7) proves the hypothesis on the arithmetic side. The shared boundary and single-phase alternatives have the fixed total costs proved in Theorem 6.3 and Corollary 8.2. The three-step modifier instead has the uniform per-step comparison (2.33), with the same original denominator. Hence a fixed coefficient for its retained signed form (2.31) enters this descent directly, with the additive constant $\log 11/16$; the exceptional 2, 3 version has the additional fixed compensation in (2.30).

For completeness, the growing-head result gives an equivalent location for the rate inside the Hardy representation. Let $K_N = \lfloor (\log N)^{3/4} \rfloor$ and use this same cutoff at N and L_N . If, for fixed constants,

$$[\mathcal{T}_{K_N}(N) - \mathcal{T}_{K_N}(L_N)]_+ \leq C(1 + \log N)^a e^{A(\log N)^{7/8}} \mathcal{B}(N^{1/6})^5, \quad (11.8)$$

then (9.11) and (9.13) imply Hypothesis 1.1 with $\rho = 7/8 < \alpha$, after enlarging its constants. Since $\mathcal{B}^5 \geq 32$, the proved head allowance is absorbed into that coefficient. Conversely the same comparison bounds the tail increment from the complete increment plus the already controlled head at L_N . The cutoffs and all mixed terms are therefore matched in both directions.

11.3 Completion of the analytic implication

Proof of Theorem 1.2. Lemma 11.1 makes $\log \mathcal{B}(X) = o(\log X)$ in all three cases, because $\alpha, \rho < 1$. For every $\varepsilon > 0$ the function $\log \mathcal{B}(X) - \varepsilon \log X$ is bounded above after including a finite initial interval. This proves $R(N) = O_\varepsilon(N^\varepsilon)$.

The nonnegative terminal in (1.3) gives $|M(N)|^2 \leq NR(N)$. Applying the energy bound with exponent 2η yields $M(x) = O_\eta(x^{1/2+\eta})$ for every $\eta > 0$, with real x covered by the floor convention.

On $\text{Re } s > 1$, absolute convergence of the Dirichlet series and partial summation give

$$\frac{1}{\zeta(s)} = \sum_{n \geq 1} \frac{\mu(n)}{n^s} = s \int_1^\infty M(x) x^{-s-1} dx. \quad (11.9)$$

Let a compact set in $\text{Re } s > 1/2$ have minimum real part $1/2 + \delta$. Choose $0 < \eta < \delta$. The integrand, including any fixed number of its s -derivatives, is dominated there by a constant times $x^{-1-\delta+\eta}(1 + \log x)^m$, which is integrable. The integral defines a holomorphic function F throughout that half-plane.

The pole of ζ at 1 is removed by $(s-1)\zeta(s)$. The identity

$$(s-1)\zeta(s)F(s) = s-1 \quad (11.10)$$

holds first on $\operatorname{Re} s > 1$, and then throughout $\operatorname{Re} s > 1/2$ by the identity theorem. At a zero ρ in that half-plane the left side would vanish while $\rho-1 \neq 0$, a contradiction. Finally the functional equation

$$\xi(s) = \xi(1-s), \quad \xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

reflects a nontrivial zero left of the critical line to one right of it [1, Section 25.4]. Hence every nontrivial zero lies on $\operatorname{Re} s = 1/2$. $\square$

Corollary 11.3 (Readouts under the same premise). *Under Hypothesis 1.1, for every $\varepsilon > 0$,*

$$\begin{aligned} \sup_{u \in \mathbb{R}} |\phi_N(u)|^2 &= O_\varepsilon(N^\varepsilon), \\ \sum_{j \leq J} |b_{j,N}|^2 &\leq 4C_\varepsilon N^\varepsilon \sin^2 \left(\frac{(2J-1)\pi}{4N+2} \right) \leq \frac{C_\varepsilon \pi^2 (2J-1)^2 N^\varepsilon}{(2N+1)^2}. \end{aligned} \quad (11.11)$$

For every $\delta > 0$,

$$\int_0^\infty e^{-2\delta u} |e^{-u/2} M(e^u)|^2 du < \infty. \quad (11.12)$$

For each fixed T , $\sup_{|\tau| \leq T} R_N(\tau)$ has the same subpower bound.

Proof. Evaluation in (4.2) has norm one. The first J terms of the mode resolution dominate $\lambda_{J,N} \sum_{j \leq J} |b_{j,N}|^2$; substitute the exact inverse eigenvalue and use $\sin x \leq x$. For (11.12), choose the terminal bound with exponent less than 2δ and integrate. The phase assertion is Theorem 4.5. $\square$

A The centered scalar comparison and its compression

This appendix preserves the alternative scalar closing route of [5, Sections 26,42,52–57]. It supplies an alternative sufficient arithmetic input; it is not an additional assumption in Theorem 1.2. Set

$$\mathcal{A}(x) = \int_1^x \frac{M(t)}{t} dt = \sum_{n \leq x} \mu(n) \log(x/n), \quad U(X) = \mathcal{A}(2X) - \mathcal{A}(X),$$

with $\mathcal{A}(x) = 0$ for $x \leq 1$. Define

$$\Theta(z) = \sum_{\substack{m \leq z \\ m \text{ odd}}} \log(z/m), \quad e_o(z) = \Theta(z) - \frac{z}{2} + \frac{\log 2}{2}, \quad Q_K(N) = \sum_{a,b \leq K} \mu(a)\mu(b)\Theta(N/(ab)). \quad (A.1)$$

The subscript on e_o distinguishes this counting remainder from the prime discrepancy in Appendix B.

A.1 The exact cutoff and its complete strip

Proposition A.1 (Hyperbola identity and strip allowance). *Let N be an integer and $L = \lfloor \sqrt{N} \rfloor \geq 2$. Then*

$$U(N/2) = 2(\log 2)M(L) - Q_L(N). \quad (A.2)$$

If $L \geq 8$, $1 \leq H \leq L/4$ is an integer, and $K = L - H$, then

$$U(N/2) = 2(\log 2)M(K) - Q_K(N) + G_K(N), \quad |G_K(N)| \leq \frac{6H^3}{L}, \quad (A.3)$$

where $G_K(N) = \sum_{a,b > K, ab \leq N} \mu(a)\mu(b) \log(N/(ab))$.

Proof. The Dirichlet identity $\mu * \mathbf{1}_{\text{odd}} = \delta_1 - \delta_2$ gives $\sum_{b \leq z} \mu(b) \Theta(z/b) = \log z - \log_+(z/2)$. The unrestricted double sum is $U(N/2)$. No pair $a, b > L$ has $ab \leq N$. Splitting the hyperbola at L , and using $N/a \geq 2$ for $a \leq L$, gives (A.2).

For the strip put $I = \{K+1, \dots, L\}$ and $F_a = \sum_{b \leq L} \mu(b) \Theta(N/(ab))$. For $a \in I$ the omitted terms have only the odd channel $m=1$; therefore

$$F_a = \log 2 - T_a, \quad T_a = \int_L^{N/a} \frac{M(t) - M(L)}{t} dt.$$

Writing $k = L - a$ and $r_a = N/a - L$, the inequalities $N - L^2 \leq 2L$ and $L \geq 4H$ give $r_a \leq 4H(k+2)/(3H+1)$. The finite sum $\sum_{k=0}^{H-1} (k+2)^2 = H(2H^2 + 9H + 13)/6$ gives $\sum_{a \in I} r_a^2 \leq 4H^3$. Since $|M(t) - M(L)| \leq t - L$, one obtains $2 \sum_{a \in I} |T_a| \leq 4H^3/L$.

Every pair in I^2 has $1 \leq N/(ab) < 3$. Put $C_I = \sum_{a \in I} \mu(a)$ and $D_I = \sum_{a \in I} \mu(a) \log(L/a)$. Its intersection contribution is $Q_{I,I} = C_I^2 \log(N/L^2) + 2C_I D_I$. Use $|C_I| \leq H$, $\log(N/L^2) \leq 2/L$ and $|D_I| \leq 2H(H-1)/(3L)$ to get $|Q_{I,I}| \leq 2H^3/L$. Finally

$$Q_L - Q_K = 2(\log 2)C_I - 2 \sum_{a \in I} \mu(a)T_a - Q_{I,I}.$$

Substitution into (A.2) cancels the linear term and proves the bound. Splitting the full hyperbola at K gives the displayed exact G_K ; its only odd channel is one because $N/(K+1)^2 < 3$. $\square$

For example, the prescribed choice $H = \min(\lfloor L^{2/3} \rfloor, \lfloor L/4 \rfloor)$ makes the entire strip cost $O(\sqrt{N})$. The full center is retained:

$$Q_K(N) = \frac{N}{2} h(K)^2 - \frac{\log 2}{2} M(K)^2 + \sum_{a,b \leq K} \mu(a) \mu(b) e_o(N/(ab)). \quad (\text{A.4})$$

This equality is simply the definition of e_o substituted into the finite form; its second term has a negative sign.

A.2 Reciprocal-preserving compression

Let $\eta = N^{-1/4}$ and use the integer nodes

$$t_0 = 1, \quad t_{j+1} = \min\{K, t_j + \max(1, \lfloor \eta t_j \rfloor)\}, \quad t_J = K.$$

For an integer a inside a cell $[t, r]$ put $\vartheta_a = (1/t - 1/a)/(1/t - 1/r)$ and distribute its Möbius coefficient with weights $1 - \vartheta_a, \vartheta_a$ to the two endpoints. A coefficient at a node is assigned there once. Let β_j be the resulting signed node masses, and set

$$\hat{Q}_K(N) = \sum_{i,j} \beta_i \beta_j \Theta(N/(t_i t_j)).$$

Theorem A.2 (Complete compressed form). *The compression has $O(N^{1/4}(1 + \log K))$ coordinates, and*

$$\sum_j \beta_j = M(K), \quad \sum_j \frac{\beta_j}{t_j} = h(K), \quad \sum_j \frac{|\beta_j|}{t_j} \leq 1 + \log K, \quad (\text{A.5})$$

$$|Q_K(N) - \hat{Q}_K(N)| \leq \sqrt{N}(1 + \log K)^2. \quad (\text{A.6})$$

In particular both centering terms in (A.4) are preserved exactly.

Proof. The weights preserve one and $1/a$ and are nonnegative. This proves (A.5). For the error, fix $b \leq K$ and put $c = N/b$. In logarithmic coordinates the kernel is $k(x) = \Theta(ce^{-x}) = \sum_{m, \text{odd}} (\log(c/m) - x)_+$. Its second derivative is the sum of unit atoms at the hinges.

On a cell $[t, r]$ put $g = r - t$, $d = \log(r/t)$ and $v = \log(a/t)$. A width-one cell moves no interior integer; otherwise $g \leq \eta t$. With log-linear weight v/d , one hinge contributes at most $d/4$ at each integer and at most $gd/4 \leq \eta^2 c/(4m)$ over the cell. Hinges occur once, and their reciprocal sum over $c/K \leq m \leq c$ is at most $1 + \log K$. The complete log-linear error is thus at most $\eta^2 c(1 + \log K)/4$.

Concavity of $1 - e^{-v}$ gives $0 \leq \vartheta_a - v/d \leq d^2/[8(1 - e^{-d})]$. The endpoint kernel difference is at most $c(1 - e^{-d})/t$, so changing to reciprocal weights costs at most $cd^2/(8t) \leq (1 + \eta)\eta^2 c/(8a)$. Because $\eta \leq 1$, the summed one-coordinate error is at most $\eta^2 N(1 + \log K)/(2b)$. Telescope the two-coordinate replacement. The original weights and compressed weights both have reciprocal absolute sum at most $1 + \log K$ by (A.5). This proves (A.6), including all cross terms.

Below $2/\eta$ there are at most $2/\eta$ integer nodes. Above it the increment is at least $\eta t/2$, so geometric growth gives the stated coordinate count. $\square$

The coordinates are exact historical means. For $j < J$ let

$$\Delta_j = 1/t_j - 1/t_{j+1}, \quad c_j = \Delta_j^{-1} \int_{t_j}^{t_{j+1}} \frac{M(u)}{u^2} du.$$

Integration by parts in each cell yields

$$\beta_0 = c_0, \quad \beta_j = c_j - c_{j-1} \quad (0 < j < J), \quad \beta_J = M(K) - c_{J-1}, \quad h(K) = \frac{M(K)}{K} + \sum_j \Delta_j c_j. \quad (\text{A.7})$$

The weighted variance identity gives

$$I_M(K) = \sum_j \Delta_j c_j^2 + V, \quad 0 \leq V \leq \frac{1}{4}. \quad (\text{A.8})$$

For its error bound, a width-one cell has zero variance. On a cell of width $g \geq 2$ the range of M is at most g , hence its variance at most $g^2/4$. The contribution is at most $g^3/(4t_j t_{j+1}) \leq \eta^2 g/4$. Summing gives $V \leq \eta^2(K-1)/4 \leq 1/4$.

Combining the two complete errors gives

$$\left| U(N/2) - [2(\log 2)M(K) - \widehat{Q}_K(N)] \right| \leq \frac{6H^3}{L} + \sqrt{N}(1 + \log K)^2. \quad (\text{A.9})$$

A.3 The positive rough-number allowance

Put

$$T(z) = \log_+ z - 2 \log_+(z/2) + \log_+(z/4), \quad F(x, y) = \sum_{P^-(n) > y} \mu(n) T(x/n), \quad P^-(1) = \infty.$$

The nonnegative tent T is supported on $1 < z < 4$. For fixed $0 < \theta < 1$ let $B_\theta(N) = -F(N, N^\theta)$ and $D_\theta(N) = -\sum_{2 < p \leq N^\theta} F(N/p, p)$.

Proposition A.3 (Fixed-roughness positive allowance). *For each fixed $0 < \theta < 1$,*

$$B_\theta(N) = [\rho_D(1/\theta - 1) + o_\theta(1)] \frac{N}{4 \log N} > 0 \quad (\text{A.10})$$

for sufficiently large N . Here $\rho_D(v) = 1$ for $0 \leq v \leq 1$ and $v\rho'_D(v) = -\rho_D(v-1)$ for $v > 1$. For $N \geq 4$ and $N^\theta \geq 2$,

$$D_\theta(N) - B_\theta(N) = U(N/2), \quad Q_K(N) = 2(\log 2)M(K) + B_\theta(N) - D_\theta(N) + G_K(N). \quad (\text{A.11})$$

Proof. Factoring by the least prime gives the exact recursion, including the unit,

$$F(x, y) = T(x) - \sum_{p > y} F(x/p, p).$$

For $x \geq 16$ and $2 \leq y < x/4$ it becomes $F(x, y) = -P(x) - \sum_{y < p \leq \sqrt{x}} F(x/p, p)$, where $P(x) = \sum_{p, \text{odd}} T(x/p)$. The prime number theorem and partial summation give

$$P(x) \sim \frac{x}{\log x} \int_{1/4}^1 T(1/t) dt = \frac{x}{4 \log x}.$$

For fixed $V > 1$, uniformly in $y \geq x^{1/V}$, $\sum_{P^-(n) > y} |\mu(n)| T(x/n) \ll_V x/\log x$. Indeed there are at most $\lfloor V \rfloor$ prime factors; choosing the largest one bounds its choices by $O_V(x/(d \log x))$, and the reciprocal sums of the remaining factors are bounded by powers of $\sum_{y < p \leq x} 1/p = O_V(1)$.

Put $u = \log x / \log y$. Induction on bounded u -ranges in the recursion, with prime partial summation, gives leading coefficient

$$f(u) = 1 \quad (1 < u \leq 2), \quad f(u) = 1 - \int_1^{u-1} \frac{f(w)}{w} dw.$$

The induction is uniform on compact subintervals away from $u = 1$. The child transition strip $1 \leq w \leq 1 + \gamma$ has $O_V(\gamma x / \log x)$ contribution by the preceding absolute bound; let $x \rightarrow \infty$ and then $\gamma \downarrow 0$. This proves $f(u) = \rho_D(u-1)$ and (A.10). The identity $v\rho_D(v) = \int_{v-1}^v \rho_D(t) dt$ for $v \geq 1$ follows from the delay equation and its value at one; a first zero would contradict that integral, proving positivity.

The least-prime partition also gives $D_\theta - B_\theta = F(N, 2)$. If $A_o(x) = \sum_{n \leq x, n, \text{odd}} \mu(n) \log(x/n)$, the parity identity gives $\mathcal{A}(x) = A_o(x) - A_o(x/2)$. Therefore $F(N, 2) = A_o(N) - 2A_o(N/2) + A_o(N/4) = U(N/2)$. Substitute this into (A.3). $\square$

The prime number theorem used here is the standard arithmetic input [1, Section 27.12]. The asymptotic is asserted only for fixed θ ; no uniform error as $\theta \rightarrow 0$ is used.

A.4 The one-sided analytic closing implication

Lemma A.4 (Mellin positivity principle). *If $g \geq 0$ is locally integrable on $[1, \infty)$ and $G(s) = \int_1^\infty g(x)x^{-s-1} dx$ has a finite real abscissa of convergence c , then G has no holomorphic extension to a neighborhood of the real point c .*

Proof. If such an extension exists, take $a = c + \delta$ sufficiently close to c that its Taylor series converges at $a - 2\delta < c$. Differentiation under the convergent integral gives $(-1)^j G^{(j)}(a) = \int g(x)(\log x)^j x^{-a-1} dx \geq 0$. Monotone convergence in the Taylor series then gives

$$\sum_{j \geq 0} \frac{(2\delta)^j}{j!} (-1)^j G^{(j)}(a) = \int_1^\infty g(x)x^{-(a-2\delta)-1} dx < \infty,$$

contradicting the abscissa. $\square$

Theorem A.5 (Conditional scalar closure). *Suppose, with the strip choice above, for every $\varepsilon > 0$,*

$$\widehat{Q}_K(N) \geq -C_\varepsilon N^{1/2+\varepsilon} \quad (\text{A.12})$$

at all sufficiently large integer cutoffs. Then all nontrivial zeta zeros lie on the critical line. Equivalently up to the proved errors, one may assume $D_\theta(N) \leq B_\theta(N) + C_\varepsilon N^{1/2+\varepsilon}$ for a fixed θ .

Proof. Equation (A.9), $|M(K)| \leq K \leq \sqrt{N}$, and the paid errors give $U(X) \ll_\varepsilon X^{1/2+\varepsilon}$ on the half-integer grid. Almost everywhere $U'(x) = (M(2x) - M(x))/x$, with absolute value at most three. The same upper bound therefore holds for all large real X . For each $q \in (1/2, 1)$, telescope $\mathcal{A}(2x) - \mathcal{A}(x)$ on dyadic scales to obtain $\mathcal{A}(x) \leq C_q x^q$ for all $x \geq 1$, after adjusting the finite base range.

Put $g(x) = C_q x^q - \mathcal{A}(x) \geq 0$. Initially on $\text{Re } s > 1$, finite partial summation followed by absolute convergence gives

$$G(s) = \int_1^\infty g(x) x^{-s-1} dx = \frac{C_q}{s-q} - \frac{1}{s^2 \zeta(s)}.$$

The right side is regular at every real $s > q$. At $s = 1$ the reciprocal zeta function has a zero. For $s > 1$, $\zeta(s) > 0$; for $0 < s < 1$ its alternating eta series is positive and $1 - 2^{1-s} < 0$, so $\zeta(s) < 0$. If the convergence abscissa of G exceeded q , this formula would extend it regularly through that real abscissa, contrary to Lemma A.4. Thus the integral is holomorphic on $\text{Re } s > q$.

The function $s^2[C_q/(s-q) - G(s)]$ is holomorphic there and agrees with $1/\zeta(s)$ on $\text{Re } s > 1$. The identity (11.10) excludes zeros right of q . Let $q \downarrow 1/2$ and use functional-equation reflection. The alternative follows from (A.11) and (A.6). $\square$

B Prime-discrepancy blocks and their complete Mellin return

This appendix consolidates the direct prime-side argument of [2, Part I, Sections 1–5]. Define $\psi(x) = \sum_{n \leq x} \Lambda(n)$ and $e_\psi(x) = \psi(x) - x$. Let F_m be the Fibonacci numbers and $x_m = p_{F_m}$, starting at an index with distinct consecutive blocks. With $c = \varphi = (1 + \sqrt{5})/2$ and $d = c - 1$, put

$$X = x_m, \quad Y = x_{m+1}, \quad Z = x_{m+2}, \quad L = Y - X, \quad b = (Z - Y)/L.$$

The prime number theorem gives $Y/X \rightarrow c$, $Z/Y \rightarrow c$ and $b \rightarrow c$. In particular $L \asymp X$, $Z \asymp X$, and $\log x_m = m \log c + o(m)$.

On $(0, L]$ define the finite signed measures

$$d\sigma(y) = \sum_{X < n \leq Y} \Lambda(n) \delta_{n-X}(dy) - dy, \quad d\tau(z) = \sum_{Y < n \leq Z} \Lambda(n) \delta_{(n-Y)/b}(dz) - b dz. \quad (\text{B.1})$$

The factor b is the Jacobian of the late block. Write

$$f(r) = \sigma((0, r]) = e_\psi(X + r) - e_\psi(X), \quad g(r) = \tau((0, r]) = e_\psi(Y + br) - e_\psi(Y),$$

and let $m_0 = e_\psi(Y) - e_\psi(X)$. Let V_f, V_g be their integrated squared deviations from their means, and put

$$P = \int y d\sigma(y), \quad Q = \int (L - z) d\tau(z).$$

Finite-measure Fubini gives $\bar{f} = m_0 - P/L$, $\bar{g} = Q/L$, with

$$P = \sum_{X < n \leq Y} \Lambda(n)(n - X) - L^2/2, \quad Q = b^{-1} \sum_{Y < n \leq Z} \Lambda(n)(Z - n) - bL^2/2. \quad (\text{B.2})$$

Proposition B.1 (Complete centered prime increment). *Let*

$$\mathcal{R}_m = \int_0^L [e_\psi(Y + br)^2 - ce_\psi(X + r)^2] dr.$$

Then

$$\mathcal{R}_m = V_g - cV_f + \frac{c}{dL}(P + Q)^2 - dL \left[e_\psi(Y) - \frac{cP + Q}{dL} \right]^2. \quad (\text{B.3})$$

All inherited means, continuous centering, and endpoint contributions are included.

Proof. Put $E = e_\psi(X)$. The two discrepancies are $E + f$ and $E + m_0 + g$. If $h_0 = \int_0^L (m_0 + g - cf) dr$, completing the square in E gives

$$\mathcal{R}_m = \int_0^L [(m_0 + g)^2 - cf^2] dr + \frac{h_0^2}{dL} - dL(E - h_0/(dL))^2.$$

The mean identities give $h_0 = -dLm_0 + cP + Q$. Centering f, g in the first two terms and using $E + m_0 = e_\psi(Y)$ gives (B.3). $\square$

For a finite signed measure q on $(0, L]$, the bridge kernel $W_L(y, z) = \min(y, z) - yz/L$ obeys

$$\iint W_L(y, z) dq(y) dq(z) = \int_0^L (F_q(r) - \bar{F}_q)^2 dr \geq 0, \quad F_q(r) = q((0, r]). \quad (\text{B.4})$$

This follows by integrating the product of the two centered indicator functions. Hence the first three terms of (B.3) have the full two-block kernel

$$\begin{aligned} H_{EE}(y, z) &= -cW_L(y, z) + \frac{c}{dL}yz, \\ H_{LL}(y, z) &= W_L(y, z) + \frac{c}{dL}(L - y)(L - z), \\ H_{EL}(y, z) &= \frac{c}{dL}y(L - z). \end{aligned} \quad (\text{B.5})$$

In particular the mixed packet pairing is exactly

$$\iint H_{EL} dq dr = \frac{c}{dL} \left(\int y dq(y) \right) \left(\int (L - z) dr(z) \right).$$

It vanishes against every nonterminal late point exactly when the early packet has zero first moment. For any exact packet decomposition, bilinearity retains every self-pairing and twice every mixed pairing. Continuous centering is assigned once; open packets stay in that sum.

The atomic diagonal in (B.5) is bounded by $O(X^2 \log^2 X)$: the kernel is $O_c(L)$ and $\Lambda(n) \leq \log Z$ for $n \leq Z$. This is an allowance for the diagonal; all signed off-diagonal and continuous terms remain in the complete expression.

Theorem B.2 (Equivalent prime-block inputs and closure). *Put*

$$B_m^\psi = X^{-2} \int_X^Y e_\psi(x)^2 dx, \quad k_m = bX^2/Y^2 \longrightarrow 1/c.$$

The following are equivalent:

- (i) $B_m^\psi = O_\varepsilon(x_m^\varepsilon)$ for every $\varepsilon > 0$;
- (ii) $[\mathcal{R}_m]_+/x_m^2 = O_\varepsilon(x_m^\varepsilon)$ for every $\varepsilon > 0$.

Either input implies the critical-line conclusion.

Proof. Changing variables in the next block gives the exact recurrence

$$B_{m+1}^\psi = k_m(cB_m^\psi + \mathcal{R}_m/X^2). \quad (\text{B.6})$$

Under (ii), this is bounded above by $a_mB_m^\psi + K_m$, with $a_m = ck_m \rightarrow 1$ and K_m subpower in x_m . For every $h > 0$, eventually $a_m \leq e^h$ and $K_m \leq C_h e^{hm}$. Iteration gives $B_m^\psi \leq C'_h(m+1)e^{hm}$; since $\log x_m = m \log c + o(m)$, allowing arbitrarily small h proves (i). Conversely, rearrange (B.6): $\mathcal{R}_m/X^2 \leq B_{m+1}^\psi/k_m$. This proves (ii) from (i), because k_m is bounded away from zero and $Y/X \rightarrow c$.

Fix $h > 0$. The geometric block sizes and (i) give

$$\int_{x_m}^{x_{m+1}} e_\psi(x)^2 x^{-2-2h} dx \leq x_m^{-2h} B_m^\psi = O_h(x_m^{-h}).$$

Summing proves $I_h := \int_1^\infty e_\psi(x)^2 x^{-2-2h} dx < \infty$. For $\sigma > 1/2 + h$, Cauchy-Schwarz then gives

$$\int_1^\infty |e_\psi(x)| x^{-\sigma-1} dx \leq I_h^{1/2} \left(\int_1^\infty x^{2h-2\sigma} dx \right)^{1/2} < \infty.$$

With spare decay on compact sets, the function $s \int_1^\infty e_\psi(x) x^{-s-1} dx$ is holomorphic on $\text{Re } s > 1/2$. On $\text{Re } s > 1$ it equals $-\zeta'(s)/\zeta(s) - s/(s-1)$ by the von Mangoldt Dirichlet series and partial summation. At a zero $\rho \neq 1$ the logarithmic derivative has a nonzero pole, contradicting this continuation. Reflection by the functional equation completes the argument. $\square$

The coefficient interface with the Möbius source is finite and exact. For any bounded weight w with the indicated finite support,

$$\sum_{X < n \leq Z} \Lambda(n) w(n) = \sum_{a \leq Z} \mu(a) \sum_{X < ak \leq Z} (\log k) w(ak). \quad (\text{B.7})$$

This is (4.21) regrouped. For a cell $(u_i, v_i]$, take w_i equal to one before u_i , $(v_i - t)/(v_i - u_i)$ inside the cell, and zero after v_i . Its centered discrepancy mean is

$$z_i = \sum_n \Lambda(n) w_i(n) - \frac{u_i + v_i}{2} = \sum_{d \leq N} \mu(d) F_i(d) - \frac{u_i + v_i}{2}, \quad F_i(t) = \sum_{k \leq N} (\log k) w_i(kt), \quad N \geq v_i.$$

The weighted transfer of Proposition 2.8 applies to these probes with their full Gram matrix. The smooth centering and the inherited prefix remain inside the centered vector; an unweighted coefficient identity is not substituted for a norm bound.

C Logical scope, normalization, and source concordance

C.1 The role of finite verification

The relevant logical statement retained from *RH on a G String* [16, Theorem 4.1] concerns the distinction between verified finite instances and a universal theorem. Let Q be Robinson arithmetic, let $T \supseteq Q$, and let V be the set of all true Σ_1^0 sentences. By the Σ_1^0 -completeness of Q , $V \subseteq \text{Th}(Q) \subseteq \text{Th}(T)$. Consequently

$$\text{Th}(T + V) = \text{Th}(T). \quad (\text{C.1})$$

The proof is that adjoining theorems as axioms changes no deductive closure. In particular, for a true universal arithmetic sentence with decidable finite instances, adjoining those instances alone gives no new theorem over T .

This is the logical motivation for stating a scale-uniform allowance and following its consequences through the horizon maps. No assumption that RH is unprovable in arithmetic or set theory enters any analytical theorem in this manuscript. The coordinate constructions and their norm identities are ordinary mathematical statements, and the quantitative growth input is Hypothesis 1.1.

C.2 Mass, normalization, and cutoff conventions

The field mass is $1/2$ and its mass squared is $1/4$. These constants belong to the specific Green metric. The corrected mass comparison in [7, Section 3] varies the source together with the operator: for $m > 0$ let

$$H_m = -\partial_u^2 + m^2, \quad J_{m,N} = \sum_{n \leq N} \mu(n) n^{-m} \delta_{\log n}.$$

Its Green function is $e^{-m|u-v|}/(2m)$, so its matched-source energy is

$$E_m(N) = \frac{1}{2m} \sum_{a,b \leq N} \frac{\mu(a)\mu(b)}{\max(a,b)^{2m}} = \int_1^N M(x)^2 x^{-2m-1} dx + \frac{M(N)^2}{2mN^{2m}}. \quad (\text{C.2})$$

The identity follows from $\max(a,b)^{-2m}/(2m) = \int_{\max(a,b)}^\infty x^{-2m-1} dx$. For a fixed source instead, changing a positive mass gives equivalent Fourier norms because the ratio of the propagators $(k^2 + m_1^2)^{-1}$ and $(k^2 + m_2^2)^{-1}$ is bounded above and below uniformly in k . These are distinct operations.

The infinite critical source is not assumed to have a bounded limiting energy. The growth conclusion (1.8) is subpower and allows divergence. The finite identities need neither simplicity of the zeta zeros nor a limiting zero-side mean-square constant.

The prime cutoff and the numerical cutoff also retain distinct meanings. For a finite prime set Q ,

$$\prod_{p \in Q} (1 - p^{-s}) = \sum_{n: p|n \Rightarrow p \in Q} \mu(n) n^{-s}$$

is a finite sum over squarefree products, including products above a chosen numerical horizon N . The mixed-cutoff object in (4.18) keeps only $n \leq N$, and the child horizon is $\lfloor N/p \rfloor$. Every arithmetic estimate in the main argument uses that numerical horizon.

C.3 Dictionary of the conserved observable

Object	Exact meaning in this manuscript
Physical charges	$\mu(n)$, read from the complete prime valuations of n .
Weighted coefficients	$c_n = \mu(n)/n$, paired with $B_N(a, b) = \min(a, b)$.
Complete energy	$R(N) = I_M(N) + M(N)^2/N$; the terminal is retained.
Modifier	Endpoint compensation with its exact residual, (2.6).
Three-step return	Final widths 3, 1 become 2, 2; the complete-energy cost on a sixfold interval is at most $2 \log 11$, (2.32).
Prime colors	Prime occupations in (4.10); insertion obeys (4.18).
Horizon	P_N , with positive pullback ZB_N and exact local inversion.

Object	Exact meaning in this manuscript
Light-plane field	$H_0\phi_N = J_N$ in $\mathcal{E} = H^1(\mathbb{R})$, with norm squared $R(N)$.
Source flow	The unitary phases generated by H_{src} ; its horizon readout is $R_N(\tau)$.
Circle return	The two-phase kernel identity (8.4), or the all-character identity (8.11).
Infinity plane	Complete derivatives of the finite Dirichlet observable or of its specified field-valued orbit.
Quadratic plane	The full outer product, retaining mixed derivatives and arithmetic charges.
Scale observable	$\delta_N = R(N) - R(\lfloor N/6 \rfloor)$ and its normalized signed trace.
Growth premise	(1.6), equivalently (10.12); one complete signed inequality.
Arithmetic conclusion	Subpower energy, followed by the locally uniform Mellin integral and zero exclusion.

C.4 Concordance of the assembled proofs

The references below identify the original analytical locations. They are source attributions, not additional premises. The main argument’s proofs are reproduced in the body in consolidated notation. Numerical tables, implementations, experimental plots, and superseded alternatives remain in the research archive and do not enter its deductions.

Present result	Existing source location
Section 2: Green metric, modifier, native budget	[3], Theorems 2.1, 4.3, 5.3, 8.1–8.2, 10.1, 11.1 and 12.1; [2], Part III-A; [5], Sections 92–95.
Sections 2.5–2.7: three-step modifier and scale return	[18], Sections 3–6 and Corollary 4.4 (older modifier); [17], Theorem 1, Corollaries 1–2 and equations (17)–(25) (later complete-energy comparison).
Section 3: long records and terminal recovery	[4], Theorems 3.2, 4.2, 5.1 and 6.1, pp. 3–5.
Section 4: field, Hamiltonian, pullback and modes	[6], Sections 1–10, pp. 3–11; corrected operator and cutoff conventions in [7].
Section 5: inherited carry and fifth-order factor	[8], Sections 2–6, pp. 3–7.
Section 6: complete projection and chain error	[9], Sections 2–6; [11], Sections 1–2.
Section 7: completed-family descent	[10], Sections 2–3, including the coprime-parent inverse and separated-copy proof.
Corollary 7.2: complete 2, 3 comparison	Direct consequence of the coprime-parent inverse and the same finite prime-insertion identity, with every child cutoff retained.
Section 8: positive domains and exact return	[12], Theorem 1.1, Corollary 1.2 and the all-winding-character theorem.
Section 9: fixed disk, full jet and growing head	[14], Sections 3–10, especially Propositions 1–2 and equations (36)–(44).
Section 10: conditioned derivative lift and signed rank two	[15], Sections 2–5. This is distinct from the Hardy-disk paper [14].

Present result	Existing source location
Section 11: coefficient law and analytic conclusion	[13], Theorem 3.1, Corollary 3.2 and Section 7; [14], Proposition 3; [15], the complete signed-trace return.
Appendix A: scalar and smaller-prime route	[5], Sections 26, 42 and 52–57, including the recovered compression and positive allowance.
Appendix B: original prime-side A/B route	[2], Part I, pp. 14–18, and the exact cell transfer in Part III-A.
Equation (C.1): finite-instance scope	[16], Theorem 4.1; its role here is the scope of finite verification.

Acknowledgment of compilation

This manuscript is an editorial consolidation of the author’s research papers, prepared with AI assistance for source organization, notation, and typesetting. The supplied analytical proofs and their hypotheses are the basis of the text. No computational experiment was rerun in preparing this edition.

References

- [1] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, Sections 18.12, 25.4 and 27.12. <https://dlmf.nist.gov/>. Accessed 6 October 2026.
- [2] C. Michaels, *Unified Proof: Running Paper*, 218-page continuing research manuscript, 2026. Part I, pp. 14–18; Part III-A; Parts XV–XVII. The final part begins on printed p. 217.
- [3] C. Michaels, *Prime DNA: The Michaels Möbius Modifier and the Michaels Dynamic DNA Sieve*, version 2, 8 September 2026, revised 29 September 2026, 23 pp.
- [4] C. Michaels, *Prime DNA: Record Heights and Historical Möbius Energy. Finite Theorems and a Conditional Terminal-Height Descent*, 5 September 2026, 10 pp.
- [5] C. Michaels research programme, *Lifetime Census and Closing Consequences*, consolidated edition through Section 106, 5 October 2026, 119 pp.
- [6] C. Michaels research programme, *The Light Plane: An Exact Horizon Map. Prime Colors, Positive Geometry, Recoverability, and the Consequences of a Governing Growth Law*, 5 October 2026, 13 pp.
- [7] C. Michaels research programme, *Horizon Map Audit*, 6 October 2026, 8 pp.
- [8] C. Michaels research programme, *Arithmophysics: Signed Prefix–Tail Comparison. Boundary Carry and Improved Lowest-Mode Estimates*, 6 October 2026, 10 pp.
- [9] C. Michaels research programme, *Arithmophysics: Orthogonal Boundary Compression*, 6 October 2026, 7 pp.
- [10] C. Michaels research programme, *Arithmophysics: Parity Width and Separated-Packet Descent*, 6 October 2026, 16 pp.
- [11] C. Michaels research programme, *Arithmophysics: Cube Moves and Additive Descent. One Shared Error Allowance for an Entire Scale Chain*, 6 October 2026, 14 pp.

- [12] C. Michaels research programme, *The Light Plane: Connes Circles and an Exact Energy Return*, 6 October 2026, 14 pp.
- [13] C. Michaels research programme, *Arithmophysics: A Constant at the Subpower Scale. Rate Comparison, Conditional Descent, and the Complete Signed Increment*, 6 October 2026, 7 pp.
- [14] C. Michaels research programme, *The Infinity and Quadratic Planes Inside the Light Plane. An Exact Energy Inclusion, Bounds for a Growing Set of Coordinates, and the Scale Coefficient*, 6 October 2026, 14 pp. Hardy-disk construction.
- [15] C. Michaels research programme, *The Infinity and Quadratic Planes in the Light Plane*, 6 October 2026, 8 pp. Normalized field-derivative construction and calibrated quadratic lift.
- [16] C. Michaels, *RH on a G String*, 5 October 2026, 9 pp. Theorem 4.1 is the finite-verification statement used in the present manuscript.
- [17] C. Michaels research continuation, *Three-Step Grouping and One-Step Boundary Return: An Unconditional Green-Energy Comparison for the Michaels Modifier*, 6 October 2026, 7 pp. The analytical Sections 1–7 are incorporated here; the numerical checks in Section 8 are not used.
- [18] C. Michaels, *The Michaels Möbius Modifier and the Michaels Dynamic DNA Sieve: Adaptive Arithmetic Blocks and Exact Green-Energy Decomposition*, 8 September 2026, 17 pp. Sections 3–6 and Corollary 4.4. Earlier edition of [3].